Internet Appendix for *Breadth, Implied Volatility, and Forecasting with Stale Information*

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This Internet Appendix carries the robustness and scope work behind Section 9 of the paper. It is an appendix in name and a companion in use: the paper states every result below and this document shows the working, so a reader who accepts the results can stop at the paper and a referee who does not can check every one of them here. Numbering continues from the paper, tables are lettered S1 onward, and every figure in both documents is produced by the code in the repository the paper names.

S0. What this appendix is for, in one page

Forty-one substantive sections, S1 through S41, plus this claim map, is enough that a reader cannot tell from the list which tests the paper leans on and which are completeness. This page says so. Each row is a claim the paper makes, the sections that carry its evidence, and the objection those sections were built to answer; every section heading also carries its own verdict, so a reader can see what a section concluded without reading it.

One convention, stated once so no citation here is ambiguous: **equations are numbered continuously from the manuscript**. The manuscript defines (1) to (12); this appendix begins at (13). A bare equation number in either document therefore refers to exactly one equation, wherever it is defined.

Table S0. The paper's claims, the evidence behind each, and the threat each piece of evidence addresses. A claim with more than one section behind it is one that was attacked from more than one direction, not one that needed more support.

claim in the paper	evidence	threat addressed
Breadth repairs a stale mark	S2, S4, S6, S8	inference, a weak benchmark, multiple testing
The measurement is not leakage	S7, S16	an overlapping target, a target normalised at the wrong date
The effect is larger in stress	S5, S10	ratio pathology when delay costs little
Timing is what makes it work	S12	a peer feed that is itself stale
Coupling says where it works	S14, S17	external validity, and leverage in eight points
Options dominate on the asset itself	S11, S13, S22, S24	a thin options block, non-linearity, mismatched horizons
The housing result is informational	S18	the index's own three-month moving average
It is not an artefact of the estimator	S20, S21	ridge dependence, a fitted surrogate generator
The apparatus fits a marked asset	S15, S23	clean lags versus periodic marking, forecast horizon
The losses are the ones claimed	S3, S25	nesting under shrinkage, a mislabelled variance-scale MSE
The data are what we say they are	S19	the one export join that cannot be checked by comparison
The secondary grids are exploratory, and the headline is not	S18, S37	the search behind the headline, and a false-discovery correction across everything reported after it
The coupling ordering is about coupling, not about when each target was measured	S17, S38	one point's leverage, and the two targets whose vendor stopped publishing in 2021

Two rows deserve flagging because their verdicts are not "survives". S4 finds a stronger domestic control than the paper's own and the rate falls from 71% to 67% against it, which is why the introduction quotes a span. S17 finds that the coupling *ordering* is robust while the fitted line through eight points is not, which is why Section S30 declines to read a threshold off it.

S1. Robustness of the measurement

$R(\delta)$ carries S_{own} in three of its four places, so every figure above is a statement about the domestic model as much as about the foreign one. That is uncomfortable, because the paper has spent its effort on the other side: Section 6.1 compresses the foreign block, Section S27 splits it, Section S28 tries it as one regressor. The control has been the same three features throughout. This section varies it, and then varies three other things that could make the headline wrong: the sample period, the number of models we tried before arriving at this one, and the gap between training and forecast. Each was run once and is reported whatever it said.

S2. Statistical inference · *Result survives*

The bootstrap intervals of Section 5 quantify sampling uncertainty in a ratio of realised out-of-sample losses, and it is tempting to dismiss them as unable to test anything at all. That would be too strong:

any interval can be inverted into a test of the values it excludes, and a reader is right to object to a claim that says otherwise. What is true is narrower, and it is what the rest of this section rests on. Those intervals are built for $R(\delta)$ rather than for the loss differential, and they carry no adjustment for the fact that our domestic model is nested inside our cross-sectional one, which is exactly the case in which the usual Diebold-Mariano comparison loses its standard distribution. Diebold has written at length about the misuse of his own test in exactly this situation (2015). So the equal-predictive-ability question is given a test of its own rather than read off a rate interval that happens to exclude zero. We use Giacomini-White (2006) as that test, and as the primary test of this paper, because it is valid for nested models and, being loss-agnostic, covers the QLIKE result as well as the squared-error one. It requires a fixed rolling estimation window, of the kind Section 4 describes. Writing L_t^{own} and L_t^{alt} for the two models' losses on test day t and $D_t = L_t^{own} - L_t^{alt}$ for the loss differential, the statistic is

$$GW = d \div SE(d), \quad SE(d)^2 = (1/n)\,[\,\gamma_0 + 2\sum_{j=1}^{L}(1 - j/(L+1))\,\gamma_j\,], \tag{13}$$

where d is the mean of D_t over the n test days, γ_j is its sample autocovariance at lag j, and the Bartlett weights $(1 - j/(L+1))$ run to $L = 40$ lags. Both parts of that sentence need defending. The HAC correction is not optional: a five-day-ahead target overlaps on four of every five days, so the raw standard error of a daily loss differential understates by roughly a factor of three. And the bandwidth is measured rather than taken from the usual automatic rule, which gives 9 lags at this sample size and truncates the kernel while the differential still carries 0.52 autocorrelation. The measurement is here rather than elsewhere: at eleven weeks the automatic bandwidth returns 0.0494 against 0.0751 at forty lags and 0.0809 at eighty, while a moving-block bootstrap, which assumes no kernel at all, returns 0.0763. The three agree once the bandwidth outlasts the dependence and the short one does not, so L is set to the block length of Section 5: both answer the same question and letting them answer it differently is what left an earlier simultaneous band studentised by one and resampled by the other. Every statistic this corrects was inflated, which is to say every correction here is against the paper's own result. Positive GW favours the model with the substitute, and we report two-sided p-values from the standard normal throughout. Clark-West (2007) serves as a second opinion on squared loss, with a qualification developed in Section S3 that is the reason it is the second opinion, not the first.

Table S1. Test statistics, HAC-adjusted with a Bartlett kernel at the bandwidth measured in this section, forty lags rather than the nine the automatic rule would give. Positive favours the cross-sectional model. Every entry is two-sided significant at 5%. The squared-loss and Clark-West rows have $p < 0.01$ at every delay; the QLIKE row runs from $p = 0.011$ at $\delta = 8$ to $p = 0.034$ at $\delta = 3$, so it clears 5% throughout and 1% nowhere. The QLIKE row was inside 1% under the nine-lag kernel and is not under the measured one, which is the correction working rather than the result weakening: a statistic divided by too small a standard error was always going to look more decisive than it was.

δ	3	5	8	13	21	34	55
Giacomini-White, squared loss	2.96	3.88	4.29	4.28	3.99	4.09	4.28
Giacomini-White, QLIKE	2.12	2.50	2.56	2.24	2.13	2.17	2.28
Clark-West	6.13	6.45	6.49	6.07	5.52	5.48	5.54

P-values are two-sided throughout. On that basis every statistic from $\delta = 3$ onward has $p \le 0.034$. The squared-loss and Clark-West rows are all below 0.01; the QLIKE row is not, running from $p = 0.011$ at

$\delta = 8$ to $p = 0.034$ at $\delta = 3$, its weakest entry. A one-sided test against the directional alternative that the cross-sectional model is better would halve each of these, putting the squared-loss and Clark-West rows under 0.002 and the whole QLIKE row under 0.02, and the alternative is directional in substance; we report the two-sided figure anyway because it is the more demanding of the two. QLIKE, the harder loss to move, is significant at the 5% level throughout and at the 1% level nowhere, and Table S1's caption says so rather than leaving the reader to infer it from the statistics.

The coverage of the normal-quantile inversion, at each bandwidth. A coverage study on a generator matched to the dependence these differentials carry finds that no bandwidth reaches the nominal level while the inversion uses normal critical values: coverage runs from 89.4% to 93.7% for a nominal 95% across the lengths tried. What the remaining gap does with length depends on how strongly the ratio is identified. Where the denominator sits ten standard errors from zero the gap stops closing once the kernel is longer than the dependence, 92.6% at both eighty and a hundred and sixty. At the two weaker strengths it is still closing at a hundred and sixty, 93.4% and 93.7%, and has not arrived. Neither reading rescues the length. The generator's dependence stops at lag forty-five, so a kernel of eighty truncates nothing; what it does is taper, weighting the covariance at lag h by one minus h over the bandwidth, and that weighting discards a share of the long-run variance at every finite bandwidth. A fraction computed against the true long-run variance therefore cannot reach one at any length, so a shortfall measured against it cannot be closed by lengthening: that reads a property of the kernel as a property of the data.

The identification strength of every set reported. Which of the three coverage figures applies to a given interval is a question about the interval, not about the procedure, and the answer is printed beside every set the paper reports: the denominator's own HAC t-statistic. Which denominator, though, depends on which ratio, and the two run opposite ways. For a substitution rate it is what the delay cost, so it is small exactly where delay does little damage; for the share of domestic error removed, which is what Table S7 inverts, it is the domestic model's own mean squared error, a level that is large in every market state. The calm cells below therefore carry a high t-statistic on the share and a substitution rate too weakly identified to quote, and both facts are consequences of the same arithmetic rather than a disagreement between two tables. It runs from 5.3 to 6.4 across the delay grid, from 2.9 to 5.9 across the market states of Table S4, and from 4.5 to 5.5 on the stressed rungs of Table S7, so the applicable figure for all of those is the middle of the three, and none of them sits at the one and a half where two sets in three come back unbounded. Two cells run the other way and should be read against a different figure: the calm rungs of Table S7 are measured on three thousand days or more and reach 11.0 and 11.2, at the ten standard errors where the set covers best, so for those the applicable figure is the strongest of the three, 94.7%. Across the whole of that ladder, four delays by seven market states, τ runs from 3.3 to 13.5, and the one cell at 3.3 is named and discarded in Section S10 rather than relied on. Below about $b = 0.01$ the bandwidth is a few dozen lags, the fixed-b limit sits close enough to normal that the two cannot be separated at any affordable number of draws, and the simulated sequence wanders rather than climbing, which is why the value used here is held to the two conditions above rather than quoted from a fitted coefficient.

Coverage among the bounded sets. One objection to any of these figures has to be met rather than asserted away. A confidence set that is the whole line covers whatever the truth is, so a procedure can raise its coverage by declining to identify anything, and this one does return unbounded sets more often than the procedure it replaces: at the weakest strength the coverage study runs, two sets in three come back unbounded against three in five before. A coverage figure that mixes those in with

the bounded sets would be flattering itself. So the study reports coverage among the bounded sets separately, and the two are the same to within two tenths of a point: 94.6% against 94.8% at the middle strength and 94.6% against 94.5% at the weakest. The reason is that most of the unbounded sets are unions of two half-lines rather than the whole line, and a union of two half-lines excludes a middle interval and can therefore miss. Declining to identify is not, on this evidence, what buys the coverage. The figures a reader quotes an interval from are the bounded ones, and they are the same figures.

What the generator's one asserted parameter is worth. The generator behind all of these figures has one parameter that was asserted rather than measured, and it is worth saying what it is worth. The numerator and the denominator share test days, so they are correlated, and the study sets that correlation to 0.50. On the real differentials it is 0.89 at eleven weeks and 0.71 averaged over the delay grid, so 0.50 is not the correlation the data have. Rerunning the procedure at 0.71 and at 0.89 moves coverage by at most 0.9 points, against a Monte Carlo standard error of 0.49 on a difference, so the assumption is not load-bearing. That is a measurement and not a reassurance: had it moved coverage by more, the generator would have had to carry the measured value.

Two bandwidths the study cannot separate. Two values of b come closest to the nominal level here and the study does not separate them. At b = 0.033 coverage is 94.8%, 94.7% and 94.3%; at b = 0.05 it is 94.7%, 94.6% and 94.6%. The largest difference between them is 0.3 of a point, against a Monte Carlo standard error of 0.5 points on a difference at four thousand replications, so the choice is not one this evidence can make and should not be presented as one it made. It is made on width, in the direction that does not flatter the result: b = 0.05 gives the wider set at all three strengths, by 0.7, 3.1 and 6.3 points of width, and where two procedures cannot be told apart the wider one is reported. A reader who prefers the narrower is looking at the figures that decide it. One caution travels with the number: b is a fraction of the sample, so 0.05 is 243 days here and 35 in the companion note's shorter window, and a study on a different length should recalibrate rather than inherit it.

A procedure with no bandwidth at all. Section 4.6 of the paper points here for it, so it belongs here rather than only in the replication output. A self-normalised statistic (Shao 2010) replaces the kernel estimate of the long-run variance with a recursive one computed from the same path, so there is no bandwidth to choose and nothing to taper. On the same generator it covers 95.5%, 96.2% and 95.3% at the three identification strengths, against 94.7%, 94.6% and 94.6% for the inversion this paper reports, so it is the better-covering procedure of the two at every strength. It is also the wider one: median widths of 0.469, 2.033 and 3.317 against 0.367, 1.512 and 2.614, between a quarter and a third wider, and it returns an unbounded set more often, 37.8% against 17.6% at the middle strength and 77.2% against 66.8% at the weakest. The readings do not change. Covering at the nominal level with a set a third wider is not a solved problem but a wider claim, and the paper reports the narrower procedure whose coverage is within half a point of nominal rather than the wider one that clears it, with this comparison here so that the choice is visible rather than implied.

Where the moving-block bootstrap is still used. Three families keep it, each for a stated reason. The effective age of Section 9 is an inverse interpolation of the delay curve, so there is no t-statistic to invert at all. Table S3's interval is on the difference between two substitution rates, which is a difference of two ratios and not itself a ratio of two means, so the inversion does not reach it; the paired bootstrap that recomputes both ratios on the same resampled days is the right instrument there and not a leftover. And the monthly and house price measurements of Tables S14, S24 and S25

are run at a different frequency on fewer than five hundred observations, where b = 0.05 would be a bandwidth of two years and the critical value would have to be recalibrated at that length rather than inherited from this one; we declined to inherit it, and say so instead of reporting a fixed-b set we had not calibrated. The rule outside the rate is Section 4.5's and is unchanged: a quantity that is not a ratio of two means keeps the instrument its own shape calls for, which is why the gross-content intervals of Section 6, the leakage interval of Section S7 and the state-space difference of Section S20 are resampled and say so.

S3. A caveat on the Clark-West column, the placebo, and what the inversion does to the intervals · *Claim narrowed*

Clark-West is the second opinion on squared loss and carries a qualification worth stating. Its correction term assumes the larger model nests the smaller exactly; a ridge with a cross-validated penalty satisfies that only approximately, because shrinkage pulls the larger model away from the nesting, and the correction can then flatter it. The diagnostic below measures how far from exact the nesting is at each delay, and the remedy imposes a single ridge penalty on both arms, selected on the restricted one. That is the fix the objection calls for rather than a different one: the correction is flattered when the two arms are shrunk by different amounts, so equalising the shrinkage closes that channel, and it does so without leaving the estimator the rest of the paper uses. It should be said exactly what the remedy buys. A common penalty reduces the departure from exact nesting rather than completely removing it. A ridge fit on the wider design is not the ridge fit on the narrower one with the extra coefficients set to zero, whatever penalty the two are given, because the penalty acts on a different number of standardised columns in each; some departure therefore survives any choice of common penalty. What the remedy removes is the differential-shrinkage part, which is the part that moves with δ and the part the diagnostic below measures. The residual part is common to both arms and small by construction, but it is not zero, and we do not claim it is. An unpenalised refit would also make the nesting exact and would answer a different question, about a model this paper never runs. The diagnostic is a ratio with no established critical value. We offer that ratio as a description, not a threshold, and lean on the common-penalty re-run instead of on where the ratio happens to fall.

Table S2. A diagnostic and a remedy for the Clark-West column, applied to this paper's own statistics. The ratio is the mean Clark-West adjustment over the magnitude of the mean loss differential; the common-penalty columns impose a single ridge penalty, selected on the domestic arm, across both.

δ	refits where the penalties differ	adjustment ratio	Clark-West	with a common penalty	Giacomini-White	with a common penalty
0	18%	1.10	0.21	0.25	−2.08	−2.04
3	49%	1.60	6.13	6.09	2.96	2.95
5	44%	1.27	6.45	6.45	3.88	3.86
13	45%	1.11	6.07	6.03	4.28	4.25
21	33%	1.03	5.52	5.52	3.99	3.99
55	34%	0.95	5.54	5.53	4.28	4.29

The diagnostic runs from 0.95 to 1.60, and only δ = 3 exceeds 1.5, the one Clark-West entry we would not lean on. The decisive column is the next one: imposing a common penalty moves no Clark-West statistic by more than 0.05, and changes no verdict at any delay. So the channel the remedy closes is present in principle and undetectable in practice here. The channel it does not close, the residual non-

nesting of two ridge fits on designs of different width, is not measured by this table and is the reason the column stays a second opinion rather than being promoted alongside Giacomini-White. It is reported with the ratio beside it instead of as a bare p-value.

The primary test is untouched, and for a reason, not by luck. The argument above turns on Clark-West being a test about *models*, so that anything changing the estimation penalty changes what its adjustment ought to be. Giacomini-White is a test about *methods* under a fixed rolling window, and a method is the whole procedure including whatever tuning happens inside that window. The validation slice is carved out of the training window, so it is part of what is being tested rather than a contaminant of a correction, and Giacomini-White makes no finite-sample correction that could be thrown off. The prediction is that its column barely moves under the remedy, and the last two columns of Table S2 are that prediction tested: no statistic moves by more than 0.06 and no verdict changes at any delay. The substitution rates, the bootstrap intervals and the simultaneous bands involve no nested-model correction at all and are unaffected.

A second concern is that we test ten delays at once. Ten pointwise 95% intervals will breach somewhere by chance, so we bootstrapped the whole vector jointly, in moving blocks of eighty days – the dependence the project measures at the longest delay the band covers, used for the resampling and for the studentising standard error alike, since a block and a kernel of unequal length answer different questions – to respect the overlap in the target (Künsch 1989), and took the largest studentised deviation across the grid. The resulting critical value is 2.53 instead of 1.96, a heavy penalty, and the effect still clears it from $\delta = 3$ to $\delta = 55$. One casualty is worth naming: the small negative reading at $\delta = 0$ does *not* survive the simultaneous band. We do not claim the cross-section is harmful when domestic data is current, and Section 6 gives the reason it is not: net of what it costs to estimate, its content there is zero. That casualty is worth using to fix a vocabulary, because three different statements get confused in this literature and this paper needs all three. *Pointwise*: an individual interval excludes zero, which is what the cells of Table 2 report. *Simultaneous*: the effect survives a band covering the whole delay grid at once, which is the 2.53 critical value above and which $\delta = 0$ fails. *Equivalence*: the effect lies inside a margin declared negligible in advance, which this paper tests nowhere and therefore never claims. A cell that fails the first is one where no incremental contribution was detected; it is not a cell where the contribution is zero, and wherever an interval still admits a contribution worth having we say what that interval permits rather than rounding it to nothing.

The third concern is that the result is an artefact of fitting a wide model to a persistent series. We tested it by shifting the entire foreign block five years through the sample. The series keep their distributions, their autocorrelation, their crises; they lose only their alignment with the target. The substitution rate falls from +71% to −8%, and is negative at every delay we checked. Whatever the model is using, it is the alignment. The residual negative is not noise around zero and should not be read as one: it is the cost of breadth from Section 6, which is all that remains once the information is removed. A placebo of this kind lands below zero by construction, and a version of it that came back at exactly zero would be the surprising result.

Finally, Section 5 described the rate as slowing, which is a claim, not an observation. Comparing $R(55)$ with earlier values: against $R(13)$ the difference is +0.078 with an interval of $[+0.029, +0.124]$, and against $R(21)$ it is +0.042 with $[+0.010, +0.078]$, both still rising. Against $R(34)$ it is +0.018 with

$[-0.007, +0.040]$, which does not reject equality. The rate is still climbing out to about seven weeks and flat thereafter.

One last property of the rate was assumed and never tested. The rate is a ratio whose denominator is what the delay costs a domestic-only forecaster, and at short delays that quantity is near zero. A ratio with a denominator that can vanish is weakly identified, and a percentile bootstrap of it reports finite endpoints whether or not they mean anything. We therefore recomputed every interval by inverting a HAC t-test after Fieller, which returns two half-lines or the whole line when the data do not pin the ratio down. At every delay from one day to eleven weeks the confidence set is an ordinary interval, and the endpoints differ from the bootstrap's by up to 8.2 percentage points: at $\delta = 55$, [54%, 83%] against [59%, 82%], where the denominator sits 5.4 standard errors from zero. That last number is worth carrying, because the coverage study of Section 4.6 of the paper is indexed by it: no cell carrying a claim in this paper is at the one and a half standard errors where two sets in three come back unbounded, and the only cells at the ten where the set covers best are the calm rungs of Table S7, which Section 4.6 of the paper sends to the strongest of its three coverage figures rather than the middle. Across the delay grid it runs from 5.3 to 6.4, and across the market states of Table S4 from 2.9 to 5.9. The objection is right in principle and is not inert. Inverted with normal critical values the two procedures agree to within 2.5 points, and that agreement is not evidence: a moving-block bootstrap at block b destroys dependence past lag b exactly as a Bartlett kernel at bandwidth b does, so at a matched length the two cannot corroborate each other, and a coverage study confirms they fail together. The set above is inverted with fixed-*b* critical values, which cover 94.6% on average over the three identification strengths where the normal-quantile procedure covers 92.7% and the bootstrap, on the same draws, runs from 90.9% at the strongest to 95.6% at the weakest – the last of those bought with an interval nearly three times the width. It is what the paper reports. None of the three reaches the nominal level at this bandwidth, and what remains once the estimation of the long-run variance is priced is the taper: on the observed loss differentials the kernel retains 85.6% of the long-run variance here, which implies 93.0% by the same arithmetic. The simulated figures therefore describe an easier problem than the one the intervals are computed on, and are quoted beside that number. Section 4.6 of the paper gives the construction, and the one delay where the ratio does not exist at all.

S4. The domestic control and the level normalisation · *Magnitude falls*

Five stronger domestic blocks were built, each a superset of the paper's: four further own lags; quarterly and annual means; leverage terms from the signed return; a second realised measure together with the LEVEL of the trailing median; and all of them at once. Each was given the same walk-forward, the same refit schedule and the same ridge, and $R(\delta)$ was recomputed inside each arm, so every rate is measured against its own control. The strongest was chosen on own-only skill averaged across delays, never on the substitution rate, the quantity under test and must not be allowed to select its own control.

Table S3. The paper's domestic block against the strongest alternative. The first two columns are out-of-sample R^2 for the domestic model alone, δ days stale. The rate is recomputed inside each arm. The interval is on the DIFFERENCE, from a paired bootstrap that recomputes both ratios, numerator and denominator, on the same resampled days.

δ	R^2 own, paper	R^2 own, +level	$R(\delta)$ paper	$R(\delta)$ +level	difference	95% interval
0	0.5188	0.5251	n/a	n/a	n/a	n/a
5	0.3637	0.3768	51%	47%	−4.1%	[−11.9, +1.7]
13	0.2332	0.2542	64%	60%	−3.6%	[−10.0, +0.5]
21	0.1478	0.1797	67%	64%	−3.4%	[−9.2, +0.1]
55	0.0072	0.0573	71%	67%	−4.1%	[−9.8, −0.7]

The control was improvable and the rate moves. A richer domestic block beats the paper's in 11 of 45 cells, and the substitution rate falls by three to four points at every delay, significantly at $\delta = 34$ and 55. Two things about that last clause. The direction is the same at all eight delays and the size barely varies, so what separates the significant delays from the rest is power rather than mechanism, and we read the column as one result rather than as two. And the list was four delays long under the nine-lag HAC kernel, which Section S2 shows truncates inside the dependence; at the measured bandwidth the table above carries, $\delta = 13$ and 21 fall just short, at $[-10.0, +0.5]$ and $[-9.2, +0.1]$. Four points is not a reversal, since 68% sits inside the $[54, 83]$ interval Table 2 already carries, but the paired test is sharper than the overlap and it excludes zero at the two longest delays, so the gap is real and the paper quotes both figures and leans on neither.

What makes that arm strong is the part worth keeping. It carries two features and they are not the same kind of thing. One is a second realised measure, a 22-day Yang-Zhang, which is simply less noisy than the 5-day one; the other is $\log M_t$, the LEVEL of the trailing median. Splitting them settles it at once. At eleven weeks the paper's block scores +0.0072, adding the second realised measure scores −0.0007, which is worse, and adding the level alone scores +0.0708. The level does the work, at nine delays out of nine, and by a margin that grows with staleness: +0.0012 of R^2 at zero delay against +0.0715 at eleven weeks.

That is a finding about this paper's own design, not about volatility. Equation (2) divides every series by its own trailing median, and that is what puts Tokyo, São Paulo and the VIX on one scale and makes a cross-section possible at all. It is also what prevents every model in the paper from seeing how volatile the world actually is. A forecaster who has lost his recent observations still knows roughly what regime he was in, and that knowledge is worth more the more he has lost, which is where the paper claims its largest effect. The comparability the design buys is paid for in domestic information, the price rises with delay, and we have not found this trade priced anywhere in the literature the paper cites. Anyone normalising a panel this way should measure what it costs before assuming it is free.

That claim has an obvious objection and it has to be answered rather than noted. The target is $\log(\sigma^2 / M_{t+5})$, M is a 252-day median and therefore very slow, so a model handed $\log M_{t-\delta}$ could lower its squared error simply by predicting the target's own DENOMINATOR. That would be arithmetic about how the target was built, not a forecast of volatility. Nothing would be inadmissible, since every feature is dated $t - \delta$, but the finding would be empty.

The test is to re-date the normaliser and see what survives. Three targets were built, differing in nothing else: the paper's, normalised at t+5; the origin-dated one of Section 4, normalised at t; and one normalised at $t - \delta$, the forecaster's own information set. The third is decisive, because its denominator is known exactly when the forecast is made, so a model cannot buy anything by predicting a quantity it has already been given, and it is also the only one of the three a delayed forecaster could actually use. Across the four positive delays the third target retains between 86% and 103% of the gain: +0.0657 against +0.0640 at eleven weeks. Re-dating the normaliser changes nothing, so the level is not being used to undo the normalisation. It is information.

Two qualifications. The gain reaches conventional significance only at $\delta = 55$, on all three targets alike, so the short-delay column shows a direction without establishing a result. And the rate correction above survives the same test: on the implementable target the level moves $R(55)$ from 73% to 69%, a shift of the same size as the 71% to 67% reported here.

A third objection reaches past the feature list to the estimator class. Every arm above hands a delayed observation to a ridge as a stale number, where a state-space model would project it forward, so the comparison could be said to test feature counts and never the class of estimator. The obvious such model was fitted on the domestic series alone: a first-order state observed with noise, with $\gamma(0) = q/(1 - \varphi^2) + r$ and $\gamma(k) = \varphi^k q/(1 - \varphi^2)$, so that $\varphi = \gamma(2)/\gamma(1)$ and the three parameters come from the first three autocovariances of the first 1,500 rows. They are then frozen, and no foreign series enters anywhere. The moments give $\varphi = 0.9265$, a half-life of 9.1 trading days, and an observation-noise variance of zero to five decimal places. That last figure is already part of the answer: with no measurement noise the Kalman gain is one and the filtered state is the observation, to 4.4×10^{-8}, so at daily frequency there is nothing for the update step to clean up.

The projection is the rest of it, and it is algebra rather than evidence. A forecaster δ days late projects by $\varphi^{\delta+h}$, which multiplies a column by one constant; the ridge standardises each column before fitting; and $(cx - c\mu)/(c\sigma) = (x - \mu)/\sigma$. The coefficient absorbs the constant and the forecast does not move. Run both arms on the same days and the largest difference across six delays is 1.8×10^{-11}, which is the 10^{-9} guard added to the standard deviation and nothing else. At a fixed delay, projecting a state forward cannot improve any scale-equivariant linear forecaster, whatever else it does for the forecaster's intuition. The objection has to enter through the state itself, if a filtered history spans something the paper's three features do not.

It does not. The paper's control is the stale level and rectangular means over 5 and 22 days; a filter weights the whole history exponentially, so the control was given both, with exponentially weighted averages at half-lives of 5, 22 and 66 days added to what it had. Six features against three, and the six-feature block is worse at every delay tested: by 0.0018 of R^2 at the origin, 0.0321 at $\delta = 21$ and 0.0520 at eleven weeks, losing by more the staler it is. Nothing was added to the information set, since every new column is a weighted average of a history the model already held. What was added was six coefficients to estimate on a training window that does not grow, which is Section S2's cost of breadth arriving inside the domestic block.

The rate then moves in the direction the objection did not anticipate, and the reason is worth stating once in general. Write the substitution rate as N/D with N the error the cross-section removes and D the error the delay adds. Raising $S_{own}(\delta)$ by e subtracts e from both, and $(N - e)/(D - e) < N/D$ whenever $N < D$, which holds here at every delay. A weak control inflates R, exactly as the objection

says. The question was only ever which control is the weak one. Measured against the exponential block, $R(\delta)$ is higher at all five delays, by up to 4.0 points, reading 74.2% at eleven weeks where the paper's control reads 71.3% and Table S8 rounds it to 71%. The alternative is the arm that inflates. What survives this section is the correction in Table S3, which comes from the level of the trailing median and not from the class of estimator. Section S12's ruler behaves the same way: rebuilt from the exponential curve it reads the headline as 4.2 days rather than 4.6, and its interval is 2% narrower as a share of the point it surrounds, so the paper keeps the reading that flatters it less.

Three attacks on the control have now been answered one at a time, and answering an objection three times is not the same as retiring it, because a fourth control can always be named. So the control is priced as a set instead. Eight domestic blocks were run through the same walk-forward: the paper's, the four supersets of Table S3, all of them at once, the exponential block above and the projected state above. At eleven weeks the substitution rate across the eight runs from 60.7% to 74.3%. The strongest by own-only skill averaged across delays is the arm carrying the level, at 0.3021 against the paper's 0.2800, and the rate inside it is 67.4%, a paired difference of -4.1% $[-10.6, -0.3]$ from the paper's 71.5%. That reproduces Table S3's four points from a second direction, on a different feature set and a different script, which is worth more than a number agreeing with itself. One bookkeeping point, because this section now quotes the paper's own control twice and the two figures differ. The state-space comparison above reads it as 71.3% and the envelope reads it as 71.5%, and Table 2 reads it as 71%. Each file builds its panel on the delay grid its own comparison needs, so each starts its test window a few days from the others. The spread is the third significant figure of a quantity whose interval is $[54, 83]$, and every comparison drawn from it is internal to the file that made it.

One rule has to be stated because the obvious choice is wrong. The lowest figure in that set is 60.7%, and it is not the conservative rate. The arm producing it scores 0.5749 own-only at the origin and -0.0644 at eleven weeks, so it is the strongest control in the set where the denominator is set and among the weakest where the numerator is. $R(\delta)$ reads the own-only curve at two points, and an arm strong at one and weak at the other moves the rate without being a better control anywhere. Quoting it would be the mirror of the error this section is answering, made in the paper's own favour, so selection stays on own-only skill and is fixed before the rates are read.

What the envelope retires is the open-ended form of the objection. The control is no longer one choice a reader takes on trust: eight are priced, the rule is stated in advance, and a ninth moves the headline only by beating all eight on own-only skill. One further check belongs here rather than in Section 7, because it is about the control rather than about the candidates. Section 7's redundancy result was established with the paper's domestic block underneath both, and a stronger block could in principle leave less room for the options market too. Re-run inside the level-carrying control on Section 7's own window, the foreign block adds nothing detectable at any of six delays, with statistics from -1.94 to -2.86, every one of them more negative than with the paper's control. Section 7 does not rest on a weak domestic model either.

S5. Temporal stability · *No drift detected; stability not established*

Section S10 conditions on market state; this asks the different question of whether the result drifts in time. Split at the midpoint, the two halves agree at every delay, 77% against 67% at eleven weeks, with overlapping intervals at all four – and overlap is the weakest form that agreement can take, since two marginal sets overlapping is not a test of their difference and these half-sample sets are too wide to test stability either way. What this section can establish is the absence of a detected change, not the

presence of stability, and the paragraph below says how that absence was established. A rolling window tells a less settled story: R(5) runs from 22% in the window ending April 2019 to 91% in the window ending April 2015, and a line through those points falls about 22 points a decade at five days and 12 at eleven weeks. Those windows overlap by 750 of 1,000 days, so the path is descriptive and no standard error is quoted for it.

The obvious explanation is composition, since the first half holds 2008, 2011 and 2015 and Section S10 has already shown the rate is higher in stress. Holding stress fixed settles it: comparing the halves within a VIX tercile, not one of twelve cells differs significantly. The drift in the path cannot be attributed to a change in how the cross-section works, and on this sample it cannot be attributed to anything.

Conditioning does something else, though, and it is the reason Section 5 quotes the rate by state.

Table S4. The substitution rate by market state, pooled over the whole window. The VIX level is taken at t−1 and the tercile boundaries are percentiles of the whole test window, so a day's bucket is decided with the rest of the sample in hand; the paragraph after this table prices what the real-time cut does instead, and says why this one is the reported one. The intervals are the fixed-b Fieller sets of Section 4.6 of the paper, which is the construction the paper quotes for the rate itself. The same twelve cells under the whole-ratio block bootstrap of Table 2 are in the replication output: the Fieller set is at least as wide in twelve of the twelve, and the two disagree at one endpoint by as much as 15 points. τ is the denominator's HAC t-statistic, how far the delay's own damage sits from zero, and it is the axis the coverage study of Section 4.6 of the paper is indexed by: it is printed here so that a reader can find the coverage figure that applies to each cell rather than assume the best one. A calm-market rate is imprecise by construction: there is little delay damage in a calm market, so the ratio's denominator is near zero.

δ	calm	middle	stressed
5	42% [−2, 121] τ 3.4	32% [1, 58] τ 5.7	65% [43, 82] τ 3.5
13	43% [−30, 109] τ 2.9	47% [26, 70] τ 5.9	74% [59, 85] τ 3.6
21	40% [−21, 92] τ 3.5	53% [32, 77] τ 5.5	77% [59, 87] τ 3.6
55	47% [−22, 94] τ 3.0	53% [28, 76] τ 4.4	81% [67, 90] τ 4.1

How the tercile is cut, and what the other cut gives. The VIX level is the one a forecaster holds at t−1, and the boundaries are percentiles of the whole test window, so a day's bucket is decided with the rest of the sample in hand. That is a choice rather than an oversight, and the reason is that a fixed boundary keeps "stressed" meaning the same thing in 2008 and in 2017 while a trailing boundary does not: on a trailing year, a VIX of 30 in 2010 is calm. But it is not a rule a forecaster could have applied on the day, and this paper described the cut as a rolling one until the alternative was measured. Part E of the conditional lab now runs both on identical forecasts, identical losses and identical days, with only the bucket a day falls in changing. On boundaries from each day's own trailing 252 sessions the stressed rate at eleven weeks reads 71% against 81%, the middle tercile 1% against 53%, and the calm tercile 84% against 47% – so the ordering the stressed column rests on does not survive the real-time cut, and inverts. The explanation is the one the fixed boundary was chosen for: a trailing tercile measures stress relative to the recent past, and the mechanism attaches to the absolute level of volatility, so the rolling "calm" bucket is full of absolutely volatile days from the years after a crisis. Two things follow and both are stated rather than chosen between. The conditional result is about the absolute level of volatility and not about relative surprise, which is a sharper statement than the paper made before. And no conditional figure in this paper may be called

real-time: the reported cut is the fixed one, every table says so now, and the real-time column above is what the word was carrying for free.

The stressed column is the one this paper can actually report: 81% at eleven weeks with an interval of [67, 90], and 81% in each half of the sample taken separately, on half-sample sets too wide to constitute a stability test and not offered as one. Before leaning on that, one objection has to be cleared. R(δ) divides by the damage the delay does, and stressed days are by definition the days on which being late costs most, so stress is where the denominator is largest, and therefore where a ratio with that denominator underneath is best behaved. A high, tight rate in stress could mean the cross-section does more work there, or merely that the ratio is measurable there. The first is a claim about markets; the second is true of any ratio and says nothing.

Taking the ratio apart settles it. Write N(δ) for the mean squared error the cross-section removes and D(δ) for the mean squared error the delay adds, both on the same days and in the target's own units, so that R = N / D.

Table S5. The substitution rate taken apart, by market state. N is the error the cross-section removes, D the error the delay adds, both mean squared error on the log-variance target. If only D moved with state, the conditional result would be arithmetic, not a finding.

δ	N calm	N stressed	N ratio	D calm	D stressed	D ratio
5	0.0279	0.1571	5.6×	0.0669	0.2435	3.6×
13	0.0336	0.3789	11.3×	0.0790	0.5108	6.5×
21	0.0402	0.5218	13.0×	0.0994	0.6750	6.8×
55	0.0570	0.8021	14.1×	0.1222	0.9914	8.1×

Both move, and the numerator moves further: averaged over delays, stress multiplies N by 11.0 and D by 6.3. The cross-section removes more error in absolute terms when markets are frightened, not merely a larger share of a larger loss, so the conditional result is a fact about markets. At eleven weeks it removes 0.8021 of squared error in stressed markets against 0.0570 in calm ones.

So the conditional reading stands, and on absolute error as well as on the ratio. The calm column is not a small number, it is an imprecise one: it spans more than a hundred points and at none of the four delays does it exclude zero. We suspected worse. A ratio whose denominator can approach zero need not have a bounded confidence set at all, and a percentile bootstrap cannot express that, so the Fieller construction of Section 4.6 of the paper was applied to all twelve cells. Every one comes back an ordinary interval, and those twelve sets are what Table S4 reports. They are not the bootstrap's intervals. The Fieller set is at least as wide in twelve of the twelve cells, and at the calm cell at eleven weeks the two differ by 15 points, [−22, 94] against the bootstrap's [−23, 79]; at the stressed cell at eleven weeks they differ by 6, [67, 90] against [73, 91]. Agreement between the two would not have been evidence in any case, for the reason Section 4.6 of the paper gives: both procedures truncate the same dependence at the same lag, so their agreement is a property of the truncation rather than evidence about either one, and pricing the estimated long-run variance separates them. The denominator never approaches zero closely enough to misbehave, so the printed intervals stand and the calm rate is imprecise without being unidentified. That is not a failure of the estimator but a property of the question: where delay costs a forecaster almost nothing, the share of that cost which the cross-section repairs has no stable value. The calm-market rate is weakly measured and economically uninformative in this sample, but the Fieller analysis does not classify it as unidentified:

every calm-state confidence set it returns is bounded, if very wide. An unconditional rate therefore averages a well-measured result with a poorly measured one, and Section 5 says so instead of quoting the average alone.

One more thing can be asked of the calm cell before it is set aside, and it should be. The paper has chosen a coordinate that fails in a third of its sample and reported the failure as a fact about markets, and a reader is entitled to suspect the coordinate instead. The effective age of Section S12 is the alternative and it is built differently: it asks how old a plain mark would have to be to forecast as badly, so it has no denominator that can vanish, and taken in mean squared error rather than in skill it carries no benchmark variance either, which is the other quantity that goes small in a calm market.

Asked that way the calm cell still does not answer. The age is 4.5 days with an interval of [0.8, 46.7], which is 9.1 times as wide relative to its own point as the stressed cell's, and 50 of 1,500 replications cannot place it on the grid at all. The non-result therefore survives the change of instrument, and that is worth more to a reader than a number would have been: where delay costs a forecaster almost nothing there is almost nothing for the cross-section to repair, and no coordinate makes a share of nothing precise. The limitation belongs to the question and not to the ratio.

The same construction pays at the other end of the ladder, where the quantity is well determined and the unit is the one a practitioner uses. In the top tercile of VIX days a fifty-five-day-old mark plus today's foreign closes forecasts like a mark 3.8 days old [1.8, 6.0]; in the top decile, 3.5 [1.2, 6.9]; in the top 5% of days, 2.3 [0.8, 4.3]. The ladder is monotone in days as well as in shares. On the days when a stale mark is most dangerous, eleven weeks of staleness plus a current cross-section is worth a mark two days old.

Conditioning on the VIX raises a fair objection, since Section 7 says that a forecaster who holds the VIX should ignore the cross-section altogether. The two are compatible, but only for a specific reader, and it is worth being exact about who. The VIX is a public index on the S&P; a holder of an illiquid asset can see it whether or not any option trades on the thing they own. Such a reader can therefore condition on market state without being able to buy the substitute that state variable makes redundant. A reader who holds options on their own asset is in Section 7's case and should stop reading at Section 8. We also note that Section S10 works with a ladder of cuts at the 33rd, 90th and 95th percentiles while this section uses terciles, and that the tercile split was chosen after that ladder had been seen.

S6. Multiple-testing adjustment · *Result survives*

Nine distinct forecasters have been fitted to this target across the project. A single test statistic on the one that won is not the right instrument for pricing that search, so the whole set is put through White's Reality Check and Hansen's SPA, with the size of both first validated by simulation under a true null of Gaussian AR(1) draws at a coefficient of 0.5, where a nominal 5% test rejects 5.5% and 6.5% of the time respectively – a generator whose dependence dies well inside the resampling length, so the figures bound the procedure's size and not its size on this panel's own dependence, with SPA rejecting more often, as Hansen predicts. Both tests resample at the dependence this project measures, L.block_for(δ), which is forty days at zero delay rising to eighty. A block shorter than the dependence tightens the null and lowers a p-value, so both that length and a mean block of ten days – which Section 4.6's own note records as shorter than the measured dependence – are run and printed side by side: at the measured length the corrected p-values are below the bootstrap's 0.0005

resolution at $\delta = 3, 5, 13$ and 55 and 0.0010 at $\delta = 21$, against below resolution at all five on the short block, so the conclusion did not rest on the length and now does not rest on the wrong one either. At $\delta = 0$ the two lengths run the other way, 0.435 against 0.450, which decides nothing because both sit far above any threshold and that delay fails the correction on either. Against the domestic benchmark, the best of the nine survives both tests at $\delta = 3, 5, 13, 21$ and 55. It fails at $\delta = 0$, where the best margin across all nine models is +0.0012 of R^2: nothing beats the benchmark there at all, so that failure is an absence of any effect to correct.

S7. Leakage from an overlapping target · *Result survives*

The target is realised variance over the next five days, so consecutive observations share four days of outcome, and a naive split would put part of the answer in the training set. Every refit here already cuts the training window at $t - \delta - h$, which removes exactly the rows whose outcome was unresolved at the forecast date: a purge, though the paper has not called it one. Removing it inflates R^2 by +0.0023 [+0.00072, +0.00400] at zero delay and +0.0019 [+0.00012, +0.00408] at three days, so it is load-bearing and any study of a five-day target that splits without one is reporting inflated numbers. Adding an embargo on top of it costs nothing detectable in any of eighteen cells, so the gap the design already leaves is sufficient.

S8. Five markets, at every delay · *Result survives*

The five-market comparison is below, across the whole delay curve rather than at one point, and it is where the timing mechanism shows most plainly.

Table S6. Substitution rate by target market at four delays, continuous target, R^2 scale. Same-day peers are the markets closing earlier on the same calendar day; every other peer contributes the previous session. Intervals are the fixed-b Fieller sets of Section 4.6 of the paper, the construction every reported rate uses; the whole-ratio bootstrap of the same differentials is in the replication output beside them.

target	close, UTC	same-day peers	R(5)	R(13)	R(21)	R(55)
S&P 500	21:00	6	51% [33, 68]	64% [48, 77]	67% [50, 79]	71% [54, 83]
FTSE 100	16:30	4	74% [53, 89]	82% [67, 92]	84% [67, 93]	88% [74, 95]
DAX	16:30	4	55% [36, 70]	69% [53, 78]	71% [56, 80]	74% [62, 82]
Hang Seng	4,755	3,522	56% [28, 79]	64% [40, 84]	65% [44, 84]	68% [48, 85]
Nikkei 225	4,692	3,459	45% [−3, 79]	54% [17, 79]	55% [14, 79]	59% [20, 80]

The ordering by same-day peers is not perfect, since the FTSE has four and recovers more than the S&P with six, but the two ends are where the mechanism predicts. The Nikkei closes before every other market in the sample, has no same-day foreign observation at all, and recovers least at every delay; the FTSE and DAX, which see all of Asia before they close, recover most. The curves are also flatter than the S&P's: for markets with early peers the rate is already near its eleven-week value within a week.

S9. Where each result holds

The same exercise with each of five indices as the target is in Table S6, on the continuous target and at every delay.

The effect appears in all five, and the Nikkei is the case that carries information. It closes before every other market in the sample, so it has no same-day foreign observation at all, since every peer reaches it a session late, and it recovers the least, 60% against 69% to 88% elsewhere. That is what the mechanism predicts and what a spurious correlation would not produce. Past that contrast we can say little: the S&P has six same-day peers and recovers less than the FTSE with four, the intervals overlap heavily, and five markets cannot resolve an ordering. Five markets also cannot show where the effect stops, and this paper no longer shows that anywhere. Section S14 adds two targets outside this group and identifies the statistic that orders the six it scores, coupling to the cross-section a target is offered; but every target it can re-run is well coupled, so what it establishes is an ordering among markets where the substitute works and not a boundary beyond which it fails. An earlier vintage of this study scored two weakly coupled targets on which the cross-section was worth less than nothing. Their exports can no longer be obtained, so no rate from them is reported here and the boundary they marked is not claimed.

One further objection deserves an answer. Rapach, Strauss and Zhou find that lagged US returns predict foreign returns while the reverse is weak (2013). That concerns returns while ours concerns volatility, and lagged returns where ours are same-day closes observed before the target's own close. We rely on both distinctions, and a reader who rejects them should discount Sections 4 and 5 accordingly.

The scope condition matters more than the ordering. Section 7 shows that where a liquid options market exists on the asset, breadth is redundant, and the S&P 500 is the most heavily optioned index in the world. So the breadth result may be most relevant to assets without such a market, the case Section 1 opens with: a private portfolio marked quarterly, a property book on an appraisal cycle, a fund reporting NAV with a lag. Nobody writes a liquid option on those. What a holder of them can see is the public cross-section, and Sections 4 and 5 say what that is worth as a function of how stale the mark has become.

We state it rather than dress it up, because it is also the weakest point of the design. We ran the experiment on a liquid asset in order to observe the truth, and the property that makes the truth observable, a deep and continuously traded market, is the same property that brings an options market with it. The result we can measure cleanly and the case we care about are therefore separated by the one feature that decides which substitute wins. Section 11 says what would close that gap.

There remains the $\delta = 0$ row of Table 2, and its history is worth recording because the reason for the caution changed. On log-scale squared error the cross-section is worse by 0.0099, interval $[-0.0190, -0.0011]$; on QLIKE it is worse by 0.0126, $[-0.0262, -0.0009]$; on squared error on the natural variance scale it is worse by 0.0035, $[-0.0307, +0.0181]$. All three point the same way and two of the three exclude zero. That is not what this section said until Section 4.4's reconstruction was corrected. On the median-normalised scale the same comparison comes out at $+0.0070$, $[-0.0427, +0.0204]$, It does not. The normalised loss weights each day's squared error by that day's trailing median, so it is a weighted loss that had been described as an unweighted one; the unweighted natural-scale loss is the third figure above, and it agrees in sign with the other two. The flip was an artefact of a weighted loss described as an unweighted one, so a monotone transformation of the target does not move the sign; the weighting did. What remains is smaller but still a reason for caution: the effect at zero delay is about one point of R^2, and the robust loss that most resembles the headline cannot distinguish it from zero. The negative is not settled enough to press a disagreement with Korkusuz (2025) and

Jayawardena et al. (2020), who find foreign volatility improves a current forecast. We decline to press one, on the narrower ground that a one-point effect at zero delay on a single target is not the basis for a disagreement, rather than on the ground that we cannot tell its sign. Table 3's decomposition still holds conditional on the log scale: whatever is there is small, bounded near one point of R^2, and mostly the bill for estimating seven coefficients.

S10. In stress and in calm · *Effect larger in stress, unresolvable in calm*

Everything above is an average over 4,862 days, and one objection is that the interesting days are not average: a rolling median assumes a stationary volatility distribution, and across a structural macro transition it should fail to adjust. The objection is testable here because the test window contains the 2008 crisis, 2011, August 2015, February 2018, COVID and the 2022 inflation shock. It is not testable on the companion note, whose window begins in November 2023 and contains no transition at all, and the note says so instead of answering it.

Two decisions shape what follows. The first is that a single split is not enough. Six named episodes at full width run to 866 days, most of them ordinary: "the 2022 inflation shock" as the whole of 2022 is 251 days. A wide window dilutes exactly what is being tested. So the definition is tightened in rungs: VIX at $t-1$ above its own trailing 67th, 90th and 95th percentile, which needs no dates and is available to a forecaster in real time, together with the named episodes at full width and as the 41 trading days centred on each window's VIX peak. The tightening was added after the wide version had been run, and we say so instead of presenting it as the original design.

The second decision matters more. Squared errors are larger when the target is more volatile, so a larger loss differential in stress can be nothing but heteroskedasticity. Every figure below is therefore the differential divided by the domestic model's own mean squared error *inside the same regime*: a share of error removed, comparable across states in a way the raw difference is not. The raw column is reported beside it, and the two tell different stories, and that is the reason for the precaution.

Table S7. What the cross-section removes from the domestic model's error, by market state, at $\delta = 55$. Rungs are nested: the top 5% of VIX days is a subset of the top 10%. The share is mean(d_t) divided by the domestic model's own mean squared error inside the same regime, a ratio of two means, so the fixed-b Fieller inversion of Section 4.6 applies to it unchanged; the sets below are that inversion, computed inside each regime. M and the critical value are recomputed at each regime's own length with $b = M/n$ held at 0.05, so M runs from 12 days on the 246-day window to 200 on the 3,996-day one. That the critical value need not be imported from the full sample is the property fixed-b buys, and the reason this table can be inverted at all: a normal quantile on a 246-day subsample would be badly sized. τ is the denominator's own t-statistic. The bootstrap column of lab27 part A is reported beside it there and discussed below.

state	test days	M	share of domestic MSE removed	95% set	τ
VIX top 5%	367	18	63.1%	[44.5, 74.2]	4.6
VIX top 10%	644	32	57.0%	[35.9, 69.1]	5.0
VIX top 33%	1,729	86	43.2%	[19.2, 56.5]	5.5
episodes, peak ±20 days	246	12	62.9%	[40.2, 78.1]	4.5
VIX bottom 67%	3,074	154	29.9%	[17.7, 40.2]	11.2
outside every episode	3,996	200	23.8%	[13.7, 32.4]	11.0

The objection is answered in the opposite direction. The contribution does not weaken in stress; it roughly doubles, and it rises monotonically as the definition tightens, running 43.2%, 57.0% and 63.1% along the VIX ladder against 29.9% on calm days. The severest rung and the calmest have disjoint sets, 44.5 against 40.2, and that disjointness was until recently the comparison the claim table graded established. It is no longer, for reasons given in the next paragraph. The gap is in any case narrower than the bootstrap gave it, 4.3 points where the bootstrap's was 8.2. The calendar route, built from dates instead of percentiles, lands on 62.9% at its narrow window against the percentile route's 63.1%, a fifth of a point apart on two cuts that share no construction. A dose-response along a mechanical ladder is stronger evidence than any single split would have been, and we would not have had it from the version of this test we ran first.

That is as far as this table can take the claim, and it is not far enough to grade one. A grade read off two of these sets failing to overlap would not do, because non-overlap of two marginal intervals is not a test of a difference: the quantity the claim is about is the contrast, and a comparison of two sets neither of which is a set for the contrast does not bound it. The second objection is worse, because it is about the instrument rather than the logic. These rungs are short. The severest holds 367 days, and the fixed-b inversion of Section 4.6 is asymptotic in n: simulated at every one of this table's own rung lengths against dependence running to $\delta + h = 60$ days, a nominal 95% set covers 57% at 246 days, 65% at 367, 75% at 644 and 85% at 866, reaching 94% only at 3,996 and 95% on the full 4,861-day window. The two sets the grade rested on are the 367-day rung and the 3,996-day one, which cover 65% and 94%: a twenty-nine point gap in reliability between two sets being read off against each other, so the earlier comparison was between instruments of unequal and unstated accuracy. The 246-day figure is quoted because the narrow calendar window is in the same table and is worse still, not because it is one of the two compared. Both objections are reasons to estimate the contrast directly, which Section 5 of the paper now does: indicator-weighted series defined on every day, so n stays at 4,861 and the contrast is one smooth function of four means whose joint long-run covariance is estimated at full length. On the share of domestic error the cross-section removes, the stressed-against-baseline contrast is +42.0 points. Two sets are available for it and both are reported, because the construction just described yields one and the ladder's own bootstrap yields the other: the delta-method fixed-b set is $[25.0, 59.0]$ and the moving-block percentile is $[25.2, 55.8]$. They agree to within a fifth of a point at the lower end and three points at the upper, which is the whole of the reassurance available when one instrument is checking another, and the rate's set below is the first of the two so that the two halves of this comparison are measured the same way. The maximum standardised contrast across the whole ladder – the statistic that prices our freedom to report the strongest rung, which this section's own tightening created – is +5.48, which no replication of four thousand reached, so its p -value is at most 0.0002 and that figure is the test's resolution rather than a measurement. The claim survives, on a proper test of the difference and with the rung selection paid for. What does not survive is the claim read on the rate rather than on the share: the same contrast computed on $\mathrm{R}(55)$ is +15.0 points with a fixed-b set of $[-7.4, 37.5]$, which spans zero, so the state cells of Table S4 describe the rate without separating it, and the claim table grades the two separately.

Three things about this table are worth stating rather than leaving to a reader who compares constructions. First, every point estimate is unchanged, because a point estimate of a ratio does not depend on how its interval is built. Only the sets moved, and all six are wider, by 1.9 to 6.4 points; ten of the twelve endpoints moved outward, which is the direction Section 4.6's coverage study says they should move. The two exceptions are the upper end of the top-33% rung, 56.5 against 57.4, and the

lower end of the narrow calendar window, 40.2 against 40.1, each a fraction of a point on a set more than thirty points wide. Second, the count of cells whose set lies entirely above zero falls from twelve of sixteen to eleven of sixteen across the four delays this ladder is run at. The cell that changes status is $\delta = 5$ on the narrow calendar window, 26.8%, which reads $[6.4, 44.7]$ under the bootstrap and $[-7.7, 46.1]$ under the inversion: 246 days with a denominator t-statistic of 3.3, the weakest identification anywhere in this appendix. We report it as no longer separated from zero and draw nothing from it. Third, τ in this table runs 4.5 to 5.5 on the stress rungs and 11.0 to 11.2 on the calm ones, so the stress rungs sit at the middle of the three strengths Section 4.6 prices coverage over and the calm rungs at the strongest, where that study's coverage is 94.7%. Across the full ladder, four delays by seven states, of which this table shows six, it runs 3.3 to 13.5, and all twenty-eight cells returned ordinary bounded intervals. The threshold below which they would not is exact rather than approximate: equation (12) of the manuscript makes the leading coefficient positive, and so the set an ordinary interval, if and only if $|\tau|$ exceeds the critical value the inversion is run at, which here is 2.11. The weakest cell on this ladder clears it by 1.2, which buys boundedness and not precision: the $\tau = 3.3$ cell above is bounded and still spans fifty-four points, and that is the pattern to expect just above the threshold rather than an anomaly.

The economic reading is the one a risk manager would want, and it is not the reading Section S28 gave. Breadth is worth most precisely when the domestic mark is stale *and* the market is moving, which is when a stale mark is most dangerous. That is a stronger claim than the paper made before, and it belongs to the cross-section itself and to no refinement of it.

S11. Implied volatility in a stressed options market · *No effect detected*

The mirror objection to Section 7 is that redundancy given VIX and VDAX assumes a well-functioning options surface, and that during a crunch breadth might stop being redundant exactly when it matters. Applying the same ladder to the increment the foreign block adds on top of the implied-volatility model, across four delays and six states, the foreign block adds significantly in none of twenty-four cells, on the stressed rungs and the calm ones alike, down to the top 5% of VIX days. Redundancy given implied volatility is not an average concealing a stressed subsample. The severest rung carries 324 days, so an effect confined to conditions more extreme than the ladder reaches would not appear here.

S12. A stale cross-section · *Scope condition supported in the examined markets*

Every number to this point was computed with the foreign block at its freshest admissible value: for each peer, the most recent close that precedes the target's. That is a deliberate choice about admissibility and it has been doing more work than the paper admits. A desk holding a stale mark is, almost by definition, a desk with a data problem, and the infrastructure that delays a domestic mark by δ days need not be the kind that delivers seven foreign closes the same evening. Nothing above says what such a desk gets. A recommendation that assumes a same-day feed without saying so is worth less than it appears, so the assumption is removed here instead of defended.

Let g be the number of days by which the foreign feed is itself late, and repeat the walk-forward of Section 5 at every pair (δ, g). The domestic block is held δ days stale as before; the peers are held g days stale on top of admissibility. The column $g = 0$ is the paper.

Table S8. Substitution rate R(δ) when the foreign block is itself g days stale. Same walk-forward and the same losses as Table 2, but not the same test days: this file builds its panel on its own (δ, g) grid and ends with 4,852 test days against Table 2's 4,862, so its own-only R^2 at eleven weeks is 0.0066 where Table 2 has 0.0072. The $g = 0$ column is therefore Table 2's comparison recomputed on a slightly different window rather than Table 2's column: it reads 67% at three weeks and 71% at eleven where Table 2 reads 67% and 71%. A caption here claimed it reproduced Table 2, which it does not and which Section S4 had already conceded two sections earlier. A negative rate means the model carrying the aged cross-section forecasts worse than the stale domestic block alone. Undefined at $\delta = 0$, where there is no delay-induced loss to recover.

δ	$g = 0$	$g = 1$	$g = 2$	$g = 3$	$g = 5$	$g = 10$
3	36%	9%	−8%	−15%	−14%	−9%
5	51%	29%	13%	−0%	−11%	−8%
13	64%	51%	42%	34%	22%	−1%
21	67%	57%	50%	44%	33%	16%
55	71%	64%	58%	53%	45%	30%

The first column is Table 2's comparison on this file's own window, within a point of it at every delay, and the rest are not near either. Ageing the foreign feed reduces skill significantly in 25 of the 30 cells tested, and in 5 of the 6 delays a single day is already enough. At $\delta = 5$ the rate falls from 51% to 29% with one day and to zero with three. At $\delta = 3$ the cross-section becomes actively harmful from $g = 2$ onwards, the honest description of a negative rate: the delayed desk would have done better ignoring it.

Shares of a loss are a poor way to deliver that, so here it is in the unit this paper uses everywhere else: the age of a domestic mark. The own-only delay curve is re-estimated on a fine grid, and each combined model is quoted as the age of the plain stale mark that forecasts exactly as well.

Table S9. Effective age. Each cell is the age of a plain domestic mark whose own-only forecast matches the model that has a δ-day-old mark plus a foreign block g days stale, by interpolation on the own-only delay curve over sixteen ages from 0 to 55 days.

δ	$g = 0$	$g = 1$	$g = 2$	$g = 3$	$g = 5$	$g = 10$
3	1.7d	2.7d	3.3d	3.7d	3.6d	3.4d
5	1.9d	3.0d	4.0d	5.0d	5.9d	5.6d
13	2.8d	4.2d	5.6d	6.8d	8.9d	13.2d
21	3.5d	5.2d	6.7d	8.1d	10.5d	14.9d
55	4.6d	6.8d	8.5d	10.0d	12.6d	19.4d

The last row states this paper's headline in one number: a fifty-five-day-old mark plus today's foreign closes forecasts like a mark 4.6 days old. The same mark with yesterday's foreign closes forecasts like one 6.8 days old, and with the feed a week behind, 12.6 days. Because this is the sentence the paper will be quoted on, it carries an interval like every other headline here, and constructing one takes some care: the map from skill to age is the inverse of a delay curve that is steep near zero and flat far out, so a fixed error in R^2 becomes a small error in age at one end and a large one at the other, and the interval cannot be obtained by pushing an R^2 interval through the curve. Resampling test days in moving blocks and re-interpolating inside each replication, recomputing the curve, the combined model and the benchmark variance on the same resampled days, gives 4.6 days [2.6, 8.2]. The one-

day and week-behind feeds give 6.8 [4.1, 11.0] and 12.6 [9.7, 19.9]. No replication placed the model at a fresh mark or off the end of the grid, so the interpolation is not being asked to extrapolate.

The same construction prices the feed itself. Differencing the ages *within* each replication, a single day of foreign staleness costs 2.1 days of domestic freshness at eleven weeks, interval [1.4, 3.2], and five days cost 8.0 [6.2, 12.7]. Both exclude zero. The marginal intervals in the previous paragraph overlap, as marginal intervals ordinarily do and is not evidence against a paired difference; the paired construction is the one that answers the question, and it agrees with the paired skill test reported above. Fitting a line along each row of Table S9 gives the same exchange rate averaged over the whole row instead of at its first step. It runs 0.12, 0.35, 1.03, 1.12 and 1.44 days at $\delta = 3$, 5, 13, 21 and 55, rising at every step. The shape is the point. Where the mark is nearly fresh the cross-section is decoration and ageing it costs almost nothing; where the mark is old the cross-section is doing the work, and then a day of the foreign feed costs about a day of the domestic one. The readers with most to gain from Sections 4 and 5 are exactly the ones whose foreign data must be current.

One reading of that decay has to be ruled out before the rest is safe. If the foreign block worked by propagation, with volatility travelling from Tokyo to New York overnight, its value would be its earliness, the decay above would be the wave passing, and the right response would be a richer dynamic model, not a warning about feeds. The cross-correlations appear to speak to this and they are a trap. Correlating each peer at $t-1$ against the target at t, then the target at $t-1$ against the peer at t, all seven peers appear to LAG the target. But a row of this panel is not an instant. The target closes at 21:00 UTC and a peer closes c hours earlier, so the first direction spans $24 + c$ hours and the second $24 - c$; for a persistent series the shorter span gives the higher correlation, and the clock alone predicts the observed sign whether or not anything leads anything.

That prediction is quantitative, so it can be tested instead of conceded. Each peer's asymmetry should scale with $2c$. Across the seven it does, with a correlation of +0.93: 0.013 and 0.019 for the FTSE and the DAX, four and a half hours ahead of New York, against 0.051 for Bovespa, a full day ahead once admissibility has shifted it. The apparent lead–lag structure is a property of the calendar, not of the markets, and it is evidence for neither story. A check from the other side agrees: replacing the seven peers with the single real-time global factor of Section 6.1 and entering it at one lag at a time, the contemporaneous factor beats every lagged version in 25 of 30 cells. Yesterday's factor does add to today's at $\delta = 3$ and $\delta = 5$, and at no longer delay, so a little genuine dynamics is present, inside an otherwise contemporaneous story.

What is left is the reading Section S27 already arrived at by a different route. The block is a second reading of one shared state, it loses value as it ages at about the rate the first reading does, and it stops contributing once it is older than the mark it was brought in to replace: at $\delta = 13$ a foreign block ten days stale returns the model to an effective age of 13.2 days, the domestic mark on its own. Breadth substitutes for a stale mark by being fresh, not by being early.

S13. An options market on another asset · *Scope condition supported in the examined markets*

Section 7 is the reason this paper's scope is narrower than its subject, and the narrowing turns on a word the section never examines. The implied-volatility block is VIX and VDAX. For an S&P target VIX is an options market written on the asset itself; for a DAX target VDAX is. For a Nikkei, Hang Seng, Nifty, ASX or Bovespa target both series belong to somebody else's index. Those are different purchases and the paper has been treating them as one.

The second case is also much closer to the case Section 1 opens with. A property book cannot buy an option on the building, because nobody writes one, and a quarterly-marked fund cannot buy an option on its own net asset value. Both can see the VIX every day for nothing, and Section S5 already leans on exactly that availability when it conditions on market state. So the question that decides how far Section 7 reaches is whether redundancy requires the option chain to be on your own asset, or whether anyone's options market will do.

It is answerable with the data already here. Each of the eight indices becomes the target in turn, with its own admissible cross-section under the clock rule of Section 4 and the same four models at each delay. Both implied-volatility series enter at $t - 1$ for every target, which is strictly admissible everywhere and identical across targets, so nothing differs between the two groups except whose options market the block contains.

Table S10. The two candidates raced against each other on every target, at $\delta = 55$. R(breadth) and R(IV) are the two substitution rates, each computed inside the same domestic control. The last two columns are what the seven foreign closes still add once implied volatility is held, and the Giacomini-White statistic testing it. Those two columns are the *seven-regressor* specification, not the compressed one the section's counts use, and the distinction matters at the DAX row: -0.0114 with $z = -1.97$ clears the pointwise 5% threshold, but it is a significantly NEGATIVE increment, which is breadth costing something rather than adding it. Three of the eight own-options cells are significantly negative in this specification, at SPX $\delta = 0$, DAX $\delta = 0$ and DAX $\delta = 55$, and none is significantly positive at any delay. Under the compressed block none of the eight reaches significance in either direction, and it is that count the text below quotes against the twenty-four. The correlation column is between the target's own decision variable and the VIX block's.

target	options market	corr. with VIX	R(IV)	R(breadth)	foreign given IV	GW z
S&P 500	its own, VIX	0.853	102.5%	71.5%	−0.0053	−0.81
DAX	its own, VDAX	0.780	104.2%	73.8%	−0.0114	−1.97
FTSE 100	another index's	0.754	85.2%	87.9%	+0.0181	+1.57
ASX 200	another index's	0.715	72.2%	69.3%	+0.0244	+1.50
Hang Seng	another index's	0.661	75.0%	67.7%	+0.0021	+0.14
Bovespa	another index's	0.659	67.6%	49.9%	−0.0201	−1.71
Nikkei 225	another index's	0.595	52.6%	59.5%	+0.0004	+0.02
Nifty 50	4,786	0.568	53.9%	53.9%	+0.0424	+1.83

The two groups do not behave alike. Where the options market is the asset's own, implied volatility wins by thirty-one points: 103% of the delay damage repaired against 73% for the cross-section, and a rate above 100% is this paper's own signal from Section 4 that the substitute is supplying something the domestic history never held. The foreign block adds significantly in none of the eight cells there, which is Section 7's redundancy result reproduced on a second target with a second exchange's index.

Where the options market belongs to another index the gap collapses to three points, 68% against 65%, and breadth is ahead outright on two of the six. The redundancy goes with it, though by less than the own-index comparison would suggest. With the HAC bandwidth set from the measured dependence rather than from the automatic rule, the foreign block adds significantly in 1 of 24 cells, and that cell is the Nifty at $\delta = 21$, at +2.29. At eleven weeks the three largest statistics are the Nifty's +1.83, the FTSE's +1.57 and the ASX's +1.50, all of which point the same way and none of which

clears the threshold. This section previously reported those three as significant, at +2.36, +2.01 and +2.13; they were divided by a standard error that truncated its kernel inside the dependence, and the claim does not survive the correction. What survives is a direction reproduced on five of six targets and not a count of significant cells, and the compressed version below is where the evidence actually sits.

One correction belongs here before those counts are read, and it changes them. Every increment above is measured with the seven-regressor block, which Sections 6.1 and S28 both say is not the model to run: seven coefficients cost one to five points of R^2, and on the S&P that cost was the whole of the negative increment given implied volatility. The bill is not the same size on every target, since it grows with delay and the eight targets differ in sample length and signal strength, so a cross-target comparison of net increments is partly a comparison of estimation costs. Giving every target the compressed block instead, the equal-weighted mean of its own admissible peers at a cost of one coefficient, removes that confound.

The split does not merely survive the correction, it sharpens. Where the options market is the asset's own, the mean increment rises from −0.0080 to +0.0029 and the count of significant cells stays at 0 of 8: compression removes the apparent harm and reveals nothing underneath it, which is Section S28's result on a second target. Where it belongs to another index, the mean increment rises from +0.0044 to +0.0153 and the count goes from 1 of 24 to **13 of 24**, with a mean Giacomini-White statistic of 1.42. Over half of the cells in which a forecaster can see an options market but cannot buy one on what they hold are cells where the cross-section still pays, and the thirteenfold jump in that count from removing six coefficients is the section's result: what looked like redundancy on this group was mostly an estimation bill.

The gradient behind the split sharpens with it. Correlating each target's decision variable with the VIX block measures how close the only options market a forecaster can see sits to the thing they hold, and against the compressed increments that closeness correlates −0.77 across the eight targets, where the seven-regressor version read −0.37. Eight points is not a regression and no interval is quoted for it, but it is no longer a relationship a reader has to squint at.

The correction also disposes of the one target that did not fit. Bovespa looked like a market holding its own options chain, with the cross-section adding nothing and doing so significantly. It was paying for six coefficients it did not need: −-0.0148 with seven regressors and +0.0098 with one. Two other explanations were available and both are ruled out rather than left open. The clock is not the cause, because entering the S&P same-day for a Bovespa forecaster, which this paper's own admissibility rule forbids and which is run only as a diagnostic, leaves the increment negative at all three delays. Nor is spanning: the partial correlation of the peer factor with the target given VIX, which is the quantity theory would reach for first, correlates −0.14 with the increment and predicts nothing. The lesson generalises past this table. A comparison of net increments across targets is partly a comparison of estimation bills, and on one target in eight that was enough to reverse the sign of the conclusion.

This does not reach the asset the paper is about, and one thing about the lower half has to be said before it is read as more than it is. Options trade on every target in it. Japan publishes the Nikkei Volatility Index, Hong Kong the VHSI, India the India VIX, Australia an S&P/ASX 200 VIX and Brazil an IVol-BR. What puts those targets in the lower half is that this paper holds VIX and VDAX and not those series, so the split is about the forecaster's information set and not about which option chains

exist. That information set is the right analogy for Section 1's reader, who can see a public volatility index and cannot buy one written on what they hold, and it is not the same claim as an asset with no options at all. Section S14 supplies targets where the absence is a property of the market, and Section 11 keeps the gap rather than claiming it is closed.

S14. Markets with no volatility index · *Scope condition supported in the examined markets*

The split above rests on what a forecaster holds rather than on what exists, so the obvious next question is whether it survives on a target where nothing exists to hold. Two are scored here for which we could find no published implied-volatility index: the IDX Composite in Indonesia and the FTSE Bursa Malaysia KLCI. That is a statement about what we could find, and a reader who knows of such an index for either market should treat that row accordingly. Two others were added in an earlier vintage of this study, the KSE-100 in Pakistan and the Colombo All-Share in Sri Lanka, and are not scored now, for the reason given below. The series are single daily files from one vendor, used in this subsection and nowhere else; no other number in this paper depends on them.

Both close at 09:00 UTC, so each sees Tokyo, Sydney and Hong Kong on the same day and Europe and New York a session late, and so do the four they are compared with, which close between 06:00 and 10:30 UTC. That clock handicap has to be held fixed rather than compared away, so the four targets of Section S13 that close in those hours are re-run through the same code, the same delays and the same implied-volatility timing. The comparison is then between markets that differ in whether anyone publishes a volatility index on them.

Table S11. Six targets with a foreign implied-volatility block, sorted by coupling: the correlation between the target's own decision variable and the equal-weighted mean of the cross-section it is offered, on its own test days. R(breadth) and R(IV) are at $\delta = 55$. The last two columns are what a compressed cross-section adds once implied volatility is held, and its Giacomini–White statistic. Two further targets, the Colombo All-Share and the KSE-100, were scored in an earlier vintage and are not scored here: their exports are no longer distributed by the source that supplied them and are not held by us, so a rate computed on them could be reported but never re-run. They were the two least coupled of the eight, so this table spans a narrower band than the earlier one and no row in it is weakly coupled.

target	volatility index	coupling	R(breadth)	R(IV)	breadth given IV	GW z
ASX 200	exists, not held	0.808	69.0%	73.0%	+0.0271	+2.45
Hang Seng	exists, not held	0.735	67.2%	74.6%	+0.0203	+1.94
Nikkei 225	exists, not held	0.704	58.4%	53.0%	+0.0292	+2.20
Nifty 50	exists, not held	0.661	53.7%	53.9%	+0.0384	+2.20
IDX Composite	none published	0.647	42.4%	38.7%	+0.0455	+2.05
FTSE Bursa KLCI	none published	0.612	44.4%	56.1%	+0.0143	+1.51

The question is answered on both of the two targets with no volatility index whose data can be re-run. Indonesia and Malaysia behave like the matched group: breadth repairs 42.4% and 44.4% of the delay damage at eleven weeks, and a compressed cross-section still adds to implied volatility there, with statistics of +2.05 and +1.51. Only the first of those clears the threshold; the Malaysian one falls short of the threshold and is reported as a direction. Both were above it under the nine-lag kernel, at +3.52 and +2.00, and neither is above it by much at the measured bandwidth, so the pair is worth no more than what two targets can carry. No options market appears anywhere in that picture, which is

the part that does not depend on a statistic: Section S13's reading does not turn on which volatility series we happened to buy.

The other two cannot be answered here, and the reason is not a result. The KSE-100 and the Colombo All-Share were scored in an earlier vintage of this study, but their exports are no longer distributed by the source that supplied them and are not held here either, so any rate computed on them could be reported and never re-run, by a reader or by us. This paper reports only what can be re-run, so they are not scored and no rate from them appears anywhere in it. The rule is verifiability and it is applied before any rate is read; it would have dropped a target whichever way that target came out. Group means for the two columns of the table are printed by the lab and set aside rather than read, because grouping and coupling are confounded across these six targets and six targets cannot separate them.

What sorts the six is not the options market, and it is visible before any forecast is run. Sorted by coupling, the table is almost monotone in the substitution rate, and across the six targets coupling and R(breadth) correlate +0.92. The lowest coupling here is Malaysia's 0.612 and the highest Australia's 0.808, so every target in the table is well coupled and none of them tests what breadth does to a market that is not. Six points are six points, and the ordering is what they carry; the mechanism behind it is the paper's own. Section S27 finds the entire recovery in the leading component of the foreign block, so a market that does not load on that component has nothing to recover with.

That converts a refusal into a rule. Section 10 declines to extrapolate the rate to a private asset, on the grounds that nothing here bounds that asset's correlation with the public cross-section. These six targets are what the refusal looks like when it is measured: coupling is the part of the story this design can see, it varies from 0.612 to 0.808 across markets that all look like ordinary equity indices, and it is computable by anyone holding a stale series and a public cross-section before they buy anything. What this paper cannot say with a number is what breadth does for a holder whose asset is weakly coupled, because the only two weakly coupled targets it ever measured are the two whose data cannot be re-obtained, and their rates are therefore not reported. Section 10's refusal to extrapolate rests on the mechanism of Section S27 and on the absence of any measurement below a coupling of 0.612, which is a narrower ground than a measured failure and is stated as the narrower one.

S15. Clean lags against periodic marking · *Approximate equivalence*

Section 10 will argue that what carries from this paper to an illiquid asset is the apparatus rather than the numbers. One part of that claim can be checked rather than hoped for, and this section checks it. Every result here models staleness as a clean lag: the forecaster sees the state exactly as it stood δ days ago and nothing since. A marked asset is not stale in that way. It is revised on a schedule and held flat in between, and the revision is a partial adjustment towards current evidence, not a full mark, which is how the property literature has modelled appraisal indices for thirty years (Geltner et al. 2003). If the substitution rate under that kind of staleness looked nothing like the rate under a clean lag, $R(\delta)$ would be a curve about clean lags alone.

It does not. Rebuilding the domestic block as a mark revised every q days and held between revisions, and matching on mean staleness, gives 64% against the clean lag's 59% at ten days, 71% against 70% at thirty-one, and 70% against 71% at ninety-four. Adding partial adjustment on top moves it a few points the other way, 63% where matching predicts 70%, so the equivalence is

approximate and we claim no more than that. Pure exponential smoothing is not staleness at all, and is reported here because it is the obvious way to model an appraisal index and the wrong one. A weighted average of the recent past still contains today's observation, so it costs almost nothing. R^2 falls from 0.5053 to 0.4780 across a mean lag of eleven weeks, against 0.4738 to +0.0072 under a clean lag, and leaves almost no damage for breadth to repair. Holding a number flat between revisions is what makes a mark stale, not blurring it.

S16. The implementable target at every delay · *Result survives*

One detail of that definition needs stating, because it looks like look-ahead and is not. The target a_{t+5} carries M_{t+5}, a median whose window includes the five days between the forecast origin and the outcome, which a forecaster at t does not know. No feature is affected, since every one is dated $t - \delta$ or earlier for the domestic block and t for the others. What changes is that the quantity being predicted is not one a practitioner could convert back to a variance forecast at the moment they make it. The implementable alternative normalises the outcome at the ORIGIN, $\log(\sigma^2_{YZ,t+5} / M_t)$, and we ran the whole experiment both ways. The two targets differ by the five-day drift in a 252-day median, which the paragraph below prices. It changes almost nothing: the substitution rate is 71% [54, 83] against 71% [54, 83] at eleven weeks, and no delay differs by more than 1.1 points. Because the difference is that small, the paper resolves the question in the practitioner's favour and quotes the origin-dated rate wherever it states a headline. The tables keep the target-dated definition, for a reason worth being explicit about: every conditional, robustness and scope result in Section 9 and everything below is computed inside that definition, and recomputing all of them to move one number by a point would change no conclusion in the paper. Table S12 gives both columns at every delay, so a reader who wants one definition throughout can read the paper in either.

Section 4.2 quotes the origin-dated rate in every headline and computes every table on the target-dated one, on the grounds that the two differ by at most 1.1 points. That is the claim, and this is the evidence for it. The two runs share features, estimator and test days exactly; only the date at which the outcome's denominator is fixed changes. The drift between them, $\log(M_{t+5} / M_t)$, has a standard deviation 3.7% of the target's own and correlates +0.52 with it, so it is neither large nor independent of what is being predicted, and the check is not vacuous.

Table S12. The substitution rate under both definitions of the target. Target-dated normalises the outcome at t + h, which is the definition every table in the paper uses; origin-dated normalises it at t, which is the definition a forecaster could act on at the moment of forecasting. Intervals are the fixed-b Fieller sets of Section 4.6 of the paper, the construction the paper quotes for a rate. The last column is the gap in percentage points, computed before the rates are rounded, which is why a row showing 51% against 50% carries a gap of −0.9.

δ	R(δ) target-dated	R(δ) origin-dated	gap
3	36% [16, 56]	35% [16, 55]	−1.1
5	51% [33, 68]	50% [33, 67]	−0.9
13	64% [48, 77]	63% [48, 76]	−0.5
21	67% [50, 79]	67% [50, 79]	−0.4
55	71% [54, 83]	71% [54, 83]	−0.3

The increments behind those ratios move the same way and by as little: the cross-section is worth +0.3657 of R^2 at eleven weeks under the target-dated outcome and +0.3677 under the origin-dated

one, on own-only skill of +0.0072 and +0.0102. The origin-dated arm is very slightly the better of the two, which is the opposite of the direction a reader suspicious of the target-dated definition would fear.

S17. How much of the coupling correlation is one point · *Ordering survives, fitted line does not*

Section 9 reports that a target's coupling to the cross-section tracks its substitution rate at +0.92 across six markets. Two different claims live in that sentence. One is an ordering claim, which a rank statistic answers. The other is a claim about a line and about where that line crosses zero, which is what a practitioner would be tempted to use as a threshold. Both are diagnosed here, and the diagnosis matters more than it would on a wider sample, because all six targets sit inside one cluster: their couplings run from 0.612 to 0.808, a band 0.196 wide. Nothing below is a new experiment: every number comes from the six pairs already in Table S11.

Two further targets, the Colombo All-Share and the KSE-100, were scored in an earlier vintage of this study and are not scored now. Their exports are no longer distributed by the source that supplied them and are not held here either, so a rate computed on them could be reported but never re-run, by a reader or by us, and this paper reports only what can be re-run. The rule is verifiability and it is applied before any rate is read; it would have dropped a target whichever way that target came out. What it costs is stated rather than worked around, because those two were the least coupled of the eight: with them gone there is no observation of a weakly coupled target anywhere in this paper, so the ordering below is an ordering among well-coupled markets and nothing here shows what happens where coupling is small.

The correlation is +0.922, with a Fisher interval of [+0.441, +0.992] on six points, and the Spearman rank correlation, which is what the ordering claim actually needs, is +0.943. The interval excludes zero and is wide, which is what six points buy. Leaving each target out in turn, the Pearson correlation never falls below +0.896 and the Spearman never below +0.900. So the ordering is not one point, and it holds inside the cluster rather than across a gap between clusters – which is the only form the question can now take, since there is no left-hand tail left to delete.

The fitted line does not survive, and on this sample it survives even less well than before. The S&P/ASX 200 sits at the top of the band and carries a leverage of 0.685 against a mean of 0.333 across six points. Removing it moves the least-squares slope from +1.469 to +1.972 and the zero-crossing from +0.315 to +0.402, a shift of nine points of coupling in the quantity a practitioner would be most tempted to use as a threshold. And the crossing now sits below every target in the sample: no target scored here couples less than +0.612, so the line is extrapolated to the crossing rather than fitted through it. Section 9 declines to call that crossing a threshold and this is why the refusal is the right one rather than a hedge.

Table S13. Leave-one-out diagnostics for the coupling relation across the six target markets whose exports can be re-obtained. 'Slope' is the least-squares slope of R(breadth) at eleven weeks on coupling with that target deleted; 'crossing' is where that line reaches zero; 'leverage' is the diagonal of the hat matrix on the full six, whose mean is 0.333. The last row is the full sample. Two further targets were scored in an earlier vintage and are no longer scored at all: their exports are no longer distributed by the source that supplied them and are not held here, so a cell computed on them could be reported but never re-run. They were the two least coupled of the eight, which is why the band this table spans is narrow and why no claim about weak coupling is made from it.

dropped	Pearson	Spearman	slope	crossing	leverage
AXJO	+0.931	+0.900	+1.972	+0.402	0.685
HSI	+0.940	+0.900	+1.354	+0.291	0.232
N225	+0.923	+0.900	+1.464	+0.315	0.170
NSEI	+0.930	+0.900	+1.515	+0.330	0.211
JKSE	+0.955	+1.000	+1.301	+0.254	0.256
KLCI	+0.896	+1.000	+1.511	+0.326	0.444
none	+0.922	+0.943	+1.469	+0.315	0.333

Six points remain six points, and none of this makes the relation better estimated than a sample of six inside one cluster can make it. What it establishes is which of the two claims that sample can carry: the ordering, yes, among well-coupled markets; the line and its crossing, no.

S18. Does the house price result come from the index's own moving average? · *Artefact explanation rejected, magnitude falls*

Each published Case-Shiller figure for month t is a three-month moving average of sales pairs closing in months $t - 2$ to t. So consecutive monthly changes are built from overlapping transaction windows, every series in the panel carries the same induced persistence, and a reader is entitled to ask whether Section 8 recovers anything but that shared construction. The objection has to be answered with a design rather than a paragraph.

Where the overlap is and is not, stated precisely, because the answer is not what the objection assumes. At the headline delay of three months the target's own stale mark covers $[t - 5, t - 3]$ and the fresh peer marks cover $[t - 2, t]$: disjoint, so no transaction month is shared between the stale mark and the cross-section replacing it. The overlap that does exist is between the peers' window and the label's, which reaches back to months $t - 1$ and t. A fresh peer mark carries conditions in those two months, and those months are inside the target's own next published change although nobody has published them for the target yet. That is what a stale mark *is*, so it is not obviously a defect; but it is also the channel through which a smoother could manufacture a rate, and argument cannot separate the two.

The design that can is to remove every shared transaction month. Read the index only every third month and the windows behind consecutive observations are disjoint, and so are the windows behind the stale mark, the peer block and the label. All three phases of the quarterly grid are run, because choosing one would leave the choice unexplained and the three are not the same sample. One quarter of relative staleness here is the same economic object as $\delta = 3$ months in Section 8. Two estimators are reported, because on a sample a quarter the size a single metro whose own delay cost

happens to be near zero dominates a mean of ratios; the pooled figure divides the average increment by the average delay cost and cannot be driven that way.

Table S14. The Section 8 design on non-overlapping quarterly changes, fourteen metros, return target. 'Mean R' is Section 8's estimator, R per metro then averaged; 'pooled R' divides the average increment by the average delay cost. Intervals resample year-long blocks common to all fourteen metros. Delay is in quarters; one quarter is one appraisal cycle.

phase	δ	S_{own}	S_{cross}	mean R	95% CI	pooled R	95% CI	metros R>0
Jan/Apr/Jul/Oct	1	0.3086	0.4272	18.8%	[−157, 260]	48.9%	[20, 74]	11 of 14
Feb/May/Aug/Nov	1	0.3206	0.4703	52.1%	[3, 91]	58.3%	[35, 79]	12 of 14
Mar/Jun/Sep/Dec	1	0.3115	0.4676	60.4%	[1, 143]	63.5%	[38, 81]	12 of 14
Jan/Apr/Jul/Oct	2	0.2474	0.4421	99.5%	[−273, 322]	64.1%	[34, 86]	13 of 14
Feb/May/Aug/Nov	2	0.2694	0.4657	64.0%	[−7, 139]	63.7%	[39, 82]	13 of 14
Mar/Jun/Sep/Dec	2	0.2599	0.4450	57.6%	[−41, 131]	62.2%	[34, 83]	13 of 14

At one quarter of staleness the pooled rate averages 57% over the three phases and spans 49% to 63% across them, against Section 8's 67.9% [43, 91] at the same economic delay, and is positive in 35 of 42 metro-phases. On the phase average twelve of the fourteen metros are positive, which under independent metros would be $p = 0.013$ by an exact sign test; the metros are not independent, sharing one national housing cycle, which is why the intervals resample blocks common to all of them and why the sign test is quoted as the most favourable reading available rather than as the evidence.

The commitment made before this design was run was positive, inside the monthly [45%, 92%], on at least ten of fourteen, and it has to be adjudicated on the estimator the band was built from. That is the mean of per-metro ratios, since the monthly 67.9% is one: the comparable quarterly figure is the mean of ratios, 43.7%, which is 1.3 points below the band, so on the comparable estimator the pre-committed test is NOT met. On the pooled estimator it is, at 56.9%. The outcome is reported as that split. It does not touch the objection the design was built for, since a manufactured rate must go to zero and on neither estimator does it; it touches the magnitude, which the quarterly design does not reproduce under the monthly estimator.

What that settles and what it does not. Under the pooled estimator all three phases exclude zero; under Section 8's own estimator only one of the three does, so the quarterly sample cannot carry the claim on its own in the form the paper states it, and no quarterly figure is offered as a second headline. What it does dispose of is the objection it was built for. The moving-average explanation is not a claim that the rate is smaller than 67.9%, it is a claim that the rate is manufactured, and a manufactured rate must go to zero when no two quantities in the design share a transaction month. It does not. It also does not stay where it was: 57% is visibly below 67.9%, which is what one would expect if a modest part of the monthly figure is the smoother, and the paper reports the gap rather than the agreement.

The de-smoothed panel answers from the other direction and produced the more interesting finding. Applying Geltner's reverse filter $r^* = (r - \rho r_{-1}) / (1 - \rho)$, with ρ estimated on each metro's first ten years and frozen so that no observation is filtered by its own future, and re-running Section 8 unchanged, one month of staleness costs −0.0012 of R^2, which is to say it costs nothing measurable,

the sign having gone the wrong way by less than a thousandth. The denominator of $R(\delta)$ has vanished, and the ratio becomes the pathology this paper's Appendix describes rather than a measurement, so no rate is quoted. The increments remain well defined and are the answer: at $\delta = 3$ the peer block still adds +0.1351 of R^2 on a panel with the moving average inverted out of it, positive on fourteen metros of fourteen. Breadth is not paid for by the smoother.

Table S15. The Section 8 design on the de-smoothed panel, return target, fourteen metros. Increments in R^2 rather than rates, because the delay cost that $R(\delta)$ divides by has collapsed. ρ averages 0.567 across metros and the reverse filter amplifies noise by $1/(1-\rho)$, which is 3.5 on average and 10.9 at Los Angeles; the level of every column below carries that amplification and only the sign is comparable with anything else in this paper.

δ (months)	S_{own}	S_{cross}	delay cost	breadth adds	metros positive
0	0.3221	0.4160	-	-	-
1	0.3234	0.4128	−0.0012	+0.0895	13 of 14
2	0.2428	0.3806	0.0794	+0.1378	14 of 14
3	0.2230	0.3581	0.0992	+0.1351	14 of 14
6	0.1496	0.3078	0.1725	+0.1582	14 of 14

The collapse of that denominator is itself worth stating, because it is a finding about the asset class and not an inconvenience. In this panel, smoothing accounts for most of what a short delay costs. Invert it and the forecaster loses little by being late, which is the same statement as saying there is less left for breadth to repair. That is a property of a repeat-sales index measured on fourteen metros, not a general law about marked assets. The appraised assets this paper is about are smoothed, so they sit on the side of that trade where staleness costs and breadth pays.

None of this touches the larger caveat Section 8 already carries. An index is not a book. The overlap question is about how the index is constructed, not about whether a real appraised portfolio behaves like a published average of other people's transactions.

S19. The seam audit in full · *No corruption detected*

One join in the index data cannot be checked by comparing the two exports, and it is the S&P's: its files abut rather than overlap, so no row exists in both. That is the target series, and a silent corruption there would move every number in the paper. Section 10 states the limitation; this section removes the words "by comparison" from it, because Cboe publishes its own VIX history and that is a different vendor on a different data path, measuring the same underlying and spanning the join.

Three failures could occur in an export and each is visible to a different instrument, so three are used. A spliced block or a gross scale error breaks the relationship $\log \sigma^2_{YZ} = \alpha + \beta \log \text{VIX}$, tested jointly on (α, β) by a Wald statistic with the same HAC correction used everywhere else here. A one-session date misalignment barely moves a five-day rolling variance and is invisible to that test, but it is glaring in the same-day relationship between the S&P's close-to-close return and the change in the exchange's index. A mis-scaled intraday range moves Yang-Zhang and leaves close-to-close variance alone, because the high and the low enter Yang-Zhang through two of its three terms and enter close-to-close variance nowhere, so their ratio is an internal check needing no auditor at all. Each is calibrated against 400 pseudo-seams drawn from the S&P's own history and against eight dates

where the S&P file has no join, which are the dates the other indices' files happen to meet and are therefore known-clean by construction.

At the join the three statistics sit at the 70th, 4th and 25th percentiles of their own nulls. The test is one-sided in what it can conclude, and the calibration shows why: several of the known-clean dates carry a large level statistic because they sit beside COVID, where the relationship between realised and implied volatility genuinely moved. A large statistic is therefore not proof of corruption, while a small one at the date the files meet is evidence that nothing happened to the data there. The power question is the one a limitation actually turns on, so it is answered by damaging the post-seam block deliberately and re-running all three: a one-session misalignment is caught, a block spliced in from the DAX is caught, and an intraday range mis-scaled by 25% is caught, while 10% is not. One corruption is invisible to all three and is named rather than left out: multiplying every post-seam price by a constant moves nothing, because all of this reads log ratios, and it is for the same reason the one corruption that could not change a number in this paper.

For VIX the concern is settled rather than bounded, because the two vendors' files can be compared directly. On the 6,747 dates both carry they are identical to the cent on 99.91% of rows and the mean absolute difference is 0.000222 index points; two rows fall outside this project's 0.5% seam tolerance, 28 November 2008 and 24 December 2012, both shortened sessions. The audit also found something we were not looking for. The retail export carries fifteen rows on dates United States equities were shut, 11 September 2001 among them: a fabricated observation rather than a discrepancy, and one no admissibility rule here could have caught, because the row is perfectly well formed. All fifteen fall outside the panel, so no model was ever shown one. VIX is Cboe's file throughout this paper, and substituting it moved no cell of Section 7 by as much as a thousandth of R^2. We report that as measurement instead of reassurance, and note that a retail export of any series can contain rows the exchange never produced.

The official file does not, however, supply the observation Section 4 actually wants, because it carries one open, high, low and close per day and no intraday snapshot at all. The 21:00 value of VIX existed and is the right regressor; it is not in any daily product, official or retail, and the lagged and same-day arms bracket it without recovering it. Differences at $\delta \leq 2$ fall inside the simultaneous bands. The variance-scale loss is dominated by a handful of crisis days, so its intervals are wide enough to be uninformative at short delays.

S20. The state-space alternative in full · *Better estimator, and what its own parameters cost*

Section 10 names the estimator as the paper's largest limitation and reports the direction of the alternative. This section gives it. The competitor is one latent global volatility factor on an AR(1), every series loading on it, and a Kalman filter handling a missing observation by dropping that row from the measurement equation. That is what a delayed domestic mark is, rather than something imposed on the data, and it is the property that makes the comparison worth making at all: the filter is not merely a better regression, it is a model of the thing under study. It is fitted two-step after Doz, Giannone and Reichlin (2011), with the state-space parameters estimated once on the initial training block and then frozen, and the filtered state fed into the same forecasting ridge on the same days. That freezing is the standard two-step practice and it is also this comparison's weakest point, so this section reports the whole comparison a second time with the parameters re-estimated as the window grows.

With the parameters held at their training-block values, it beats the specification used throughout this paper at $\delta = 55$ by 0.0197 of R^2, interval $[+0.00169, +0.03857]$. At $\delta = 21$ it is ahead by 0.0120, interval $[+0.00004, +0.02395]$ – a set that clears zero by four hundred-thousandths of R^2, which counts as a significant win under the inversion of Section 4.6 and carries nothing beyond its sign. It loses at $\delta = 0$, by 0.0225, interval $[-0.03331, -0.00968]$, and neither model wins at three, five or thirteen days. The crossover is the informative part: imposing a one-factor structure costs skill when the mark is fresh and pays when it is stale, which is what a structural assumption should do when the data thin out. On the point estimates the sign turns between five and thirteen days; on the intervals the regression's advantage is established at $\delta = 0$ alone and the filter's from three weeks on. Its reconstruction of the current domestic state the forecaster is not allowed to see correlates 0.84 with the truth at eleven weeks, against 0.18 for the stale value itself.

What the result establishes is the size of what this paper's transparency costs: roughly two points of R^2 at the longest delay, which is larger than several of the effects the paper argues over. A bound in R^2 is not what a reader of this paper wants, so the headline itself was recomputed inside the filtered specification. The control cannot change, because Section S4 finds no filtered own-only model to build: with no observation noise in the domestic series the filtered state *is* the observation. So the filter enters through the model carrying the cross-section, where, with the parameters still frozen, it is the better model at three of six delays, and the rate at eleven weeks becomes 75.3% against the published 71.5%, a paired difference of +3.8% [+0.4%, +8.5%]. In days, the mark reads 3.7 days old [2.0, 6.8] rather than the published 4.6, a paired difference of −0.9 days [−2.3, −0.1]. The interval on 3.7 is that file's own bootstrap and is not the one Section S12 quotes beside 4.6, which is why the paired difference rather than the pair of intervals is the number to read.

Everything above holds the state-space parameters at values no forecaster had on the test days, and a filtered state computed from estimated parameters treated as known reports a smaller error than it has earned (Rodríguez and Ruiz, 2012). Re-estimating the state space every 252 rows on everything strictly before each forecast's own training cut gives twenty-one vintages, with φ running from 0.9631 to 0.9758 and the target's loading from 0.807 to 0.897. Under them the filter is the better cross-sectional model at two of six delays rather than three, and the rate at eleven weeks reads 72.6%, against 75.3% frozen and 71.5% as published. It is still ahead of the ridge at that delay on the point estimate, 0.3788 of R^2 against 0.3729, and the difference is $+0.0059$ $[-0.0124, +0.0271]$, which does not exclude zero. Two further readings say the frozen arm also understates its own uncertainty. The spread of the reconstruction across parameter draws is 0.2127 against a filtered posterior standard deviation of 0.2273 averaged over the test days, so the two together are 1.38 times the band the frozen filter reports; the draws are simulated from the fitted model and refitted rather than resampled in moving blocks, because the factor is persistent and the joins between blocks break the persistence the parameter is measuring. And standardised one-step innovations fail unit variance or whiteness on eight of eight series, the target's variance reading 1.741, which makes even the wider band a floor rather than an estimate. The direction of this section therefore survives the parameters and its margin does not, which is a reason to quote the published figure rather than the filtered one. This arm is a robustness check on an alternative the paper declares, not one of the secondary grids Section S37 pools, and it is not counted there.

The estimator limitation therefore does not run the way a limitation usually does, and that is true of the point estimates whether the filter's parameters are frozen or live. A better estimator raises the numerator of $R(\delta)$ and has nothing to give its denominator, so the paper is not protecting a large

number with a transparent estimator; it is quoting a smaller one than the better estimator would give it. What is still declined is the rest of the paper inside a filtered specification, which is a different paper: $R(\delta)$ is a ratio in which the same domestic control appears three times, so a better estimator raises numerator and denominator together and the rate is not simply the rate plus the improvement.

S21. The estimation bill under an unfitted generator · *Objection answered*

Both measurements of the bill are relative to something. The slope method extrapolates from peer counts; the surrogate method replaces the foreign block with fitted Gaussian $AR(1)$ series. One objection to the second is correct: it preserves first-order persistence and contemporaneous cross-market covariance and nothing else: not heavy tails, not volatility-of-volatility, not crisis clustering, not higher-order serial structure. Under that objection the cost is identified only relative to a generator we chose, and a forecaster's real regressors are not Gaussian $AR(1)$ series.

The objection has a direct test. Measure the bill again with a generator that fits nothing at all: take the real foreign block and move it circularly through the sample. Every marginal, every tail, every cluster is then the real one, and the only thing destroyed is the alignment with the target. If an unfitted generator returns the fitted generator's number, the objection is answered, not conceded.

Table S16. The cost of breadth under a fitted and an unfitted null, on identical days. $AR(1)$ is the generator used above, twenty draws; SHIFT is the real foreign block moved circularly with its real missing-data mask held fixed, twenty offsets, with the Monte Carlo standard error of that mean beside it. NAIVE is the same shift applied to the raw array, so the holidays move with the values; it is reported as a diagnostic, not as a candidate.

δ	net ΔR^2	cost, $AR(1)$	cost, SHIFT	(s.e.)	cost, NAIVE	gross, $AR(1)$	gross, SHIFT
0	−0.0099	−0.0120	−0.0117	0.0008	−0.0315	+0.0021	+0.0018
1	−0.0005	−0.0142	−0.0134	0.0010	−0.0317	+0.0137	+0.0128
3	+0.0398	−0.0190	−0.0168	0.0012	−0.0327	+0.0588	+0.0566
5	+0.0797	−0.0233	−0.0200	0.0015	−0.0339	+0.1029	+0.0997
13	+0.1818	−0.0393	−0.0334	0.0026	−0.0416	+0.2211	+0.2151
21	+0.2495	−0.0524	−0.0467	0.0038	−0.0497	+0.3019	+0.2962
55	+0.3657	−0.0674	−0.0667	0.0070	−0.0654	+0.4331	+0.4323

One bookkeeping point before the result, because the table invites a fair objection. The $AR(1)$ column here is twenty draws on this subsection's delay grid, not the one hundred behind Table 3, so the two tables give slightly different figures for the same quantity: -0.0120 against -0.0126 at $\delta = 0$, and -0.0524 against -0.0514 at $\delta = 21$. That spread is Monte Carlo error and nothing else. At three weeks it is 0.0010 against Table 3's own Monte Carlo standard error of 0.0014, so it is inside that error; at zero delay it is 0.0006 against 0.0003, so it is about twice it, and we say so rather than claiming the two agree everywhere. Table 3 remains the paper's estimate of the bill, because it rests on more draws. What Table S16 is for is the comparison *within* it, between two columns computed on identical days from an identical number of draws, and that comparison is unaffected by how many draws either column uses.

The two generators agree at every delay. Compared against three times its own combined Monte Carlo standard error, delay by delay rather than against the loosest tolerance in the table, the largest gap is 0.0059 of R^2 at $\delta = 13$, and the closest call against its own tolerance is at $\delta = 5$, at 0.64 of it; no

delay comes closer. The gross readings move by exactly as much, since the net reading is common to both columns and gross is net less cost: +0.0021 becomes +0.0018 at $\delta = 0$, and +0.3019 becomes +0.2962 at $\delta = 21$. Nothing in Section 6 depends on which generator supplied the number, and in that sense the objection is answered. That claim is worth only as much as the null generator behind it, and two properties of the shift are what make it a placebo rather than a plausible number with no outward sign of trouble.

The shift has to preserve the block's pattern of missing observations: one that does not quietly discards about 187 test days per draw, or 3.8%, from every comparison. And the offsets have to be guarded away from the ends, since a draw landing eight days from the identity scored +0.1818 on its own against a spread whose standard deviation is 0.0680, enough to drag an eight-draw mean. With both in place, the naive version of the penalty differs from the corrected estimate by 0.0199 of R^2 at $\delta = 0$, where the real cost is small, and by 0.0013 at $\delta = 55$, where it is large, which is the signature of a fixed charge for lost days rather than a cost of breadth. The shift is now guarded and the mask preserved.

S22. Redundancy under a non-linear estimator · *No effect detected*

Redundancy was established with a ridge, and it is reasonable to ask whether a linear map is simply too blunt to find what the foreign block holds given implied volatility. The test is to give the block a richer map and see whether anything appears: a random Fourier feature expansion, which is a one-hidden-layer network whose first layer is drawn, not learned, at two widths, and a k-nearest-neighbour arm that imposes no functional form at all.

Table S17. What each non-linear map of the foreign block adds to a forecaster holding stale domestic data and implied volatility. Wider penalty grid; Giacomini-White statistics, the implied-volatility model being nested in each. A statistic above +1.96 would overturn the redundancy result. Cells in this grid are part of the post-selection audit of Section S37; those significant only at the uncorrected 5% level are *exploratory* and no claim in this paper rests on one of them alone.

δ	R^2 IV	+ foreign	GW z	+ FACTOR	GW z	+ RFF(20)	GW z	+ RFF(60)	GW z
0	0.5507	−0.0093	−2.24	−0.0107	−2.40	−0.0400	−2.49	−0.0525	−3.40
3	0.5243	−0.0049	−0.91	−0.0095	−1.65	−0.0445	−2.01	−0.0595	−2.81
5	0.5229	−0.0055	−0.90	−0.0074	−1.18	−0.0527	−1.95	−0.0632	−2.80
13	0.5134	−0.0048	−0.65	−0.0093	−0.72	−0.0629	−2.05	−0.0720	−2.52
21	0.5124	−0.0049	−0.69	−0.0154	−1.03	−0.0679	−2.05	−0.0738	−2.58
55	0.5082	−0.0046	−0.70	−0.0047	−0.69	−0.0587	−1.99	−0.0657	−2.32

No arm adds anything at any delay, and the two random-feature arms are significantly *worse* at six of six delays for RFF(60) and five of six for RFF(20). The exception is RFF(20) at five days, which reads $z = -1.95$ and $p = 0.051$ and so falls just outside a two-sided 5% threshold. The count is taken from the lab rather than from the rounding. The direction of the capacity comparison is the informative part: RFF(60) is below RFF(20) at every delay, by 0.0059 to 0.0150 of R^2. If the objection were right, if there really were cross-market structure a linear map could not reach, more flexibility would find more of it. More flexibility finds less, and that is what over-parameterisation looks like when the signal is not there. The surrogate measurement puts a number on the mechanism: bending the

foreign block costs 0.005 to 0.008 of R^2 for the FACTOR arm and 0.017 to 0.056 for the random-feature arms, rising with both delay and width.

One thing genuinely changes under the wider grid, and it favours the objection. The *linear* increment, significantly negative at short delays in Table S22, stops being significant from $\delta = 3$ onward, with statistics between -0.65 and -0.91. That is the right result and the one Section 6's estimation-cost argument predicted: the foreign block given implied volatility is not harmful, it is empty, and the apparent harm in Table S22 was the bill for estimating regressors that carry nothing. The redundancy conclusion is unchanged; its sign is better explained.

What this does not license is the claim that no non-linear model could find anything. The random first layer is not learned, and a learned one has strictly more free parameters, so it faces the same estimation-cost bill this sample has repeatedly failed to pay, which is an argument by extension rather than a test. Section 10 records it as such.

S23. Sensitivity to the forecast horizon · *Result survives in the middle of the range*

Every number above forecasts five trading days ahead, and a reader is entitled to ask whether that is a finding or a convention. So the whole experiment is re-run at one day, two weeks and one month. One detail has to be settled before the table can be read: the Yang-Zhang proxy spans five days, which is how the latent variance is measured and has nothing to do with how far ahead we forecast, but it means that at $h = 1$ the outcome window shares four of its five days with the domestic feature the model already holds. Nothing leaks, since the outcome is still strictly in the future and every feature is still dated $t - \delta$ or earlier for the domestic block. But a forecaster at one day is being asked to predict an object they can largely already see, and their R^2 at $\delta = 0$ duly comes out at 0.924 against 0.519 at five days. The serial correlation of the target confirms it: 0.953 at one day against 0.692 at five, 0.576 at ten and 0.417 at twenty-one.

Table S18. The substitution rate at eleven weeks under four forecast horizons. Intervals are the fixed-b Fieller sets of Section 4.6 of the paper, which is why the five-day row reproduces the headline's endpoints and not merely its point estimate; the full-ratio bootstrap of Section 5 is beside them in the replication output, in moving blocks of the length Section S2 measures, held at that length across horizons so the four rows are comparable. The $h = 5$ row is the paper's headline and reproduces it exactly.

horizon	own R^2 at $\delta = 0$	R(55) under R^2	95% CI	R(55) under QLIKE
1 day	0.924	70%	[59, 77]	83%
5 days	0.519	71%	[54, 83]	81%
10 days	0.367	66%	[38, 83]	73%
21 days	0.202	50%	[8, 78]	65%

From one day to two weeks the rate is flat at 70%, 71% and 66% across a tenfold change in horizon and despite the one-day column measuring something partly mechanical. At one month it falls to 50%, the only point estimate outside the interval the headline quotes, though the intervals still overlap heavily and the one-month interval is wide enough on its own to be uninformative. QLIKE tells the same story more smoothly: 83%, 81%, 73%, 65%. So the headline is a property of the information over horizons out to a fortnight and begins to decay beyond that, which is what should happen, since a foreign close carries information about the next week's volatility state more readily than about the next month's, and by twenty-one days the domestic model it is being compared

against has itself fallen to an R^2 of 0.202, so there is less loss left to recover and a noisier denominator to recover it against. We report the five-day figure as the headline because it is the horizon the whole paper is built on, not because it is the most favourable: at one day under QLIKE the rate is higher.

S24. The race with matched horizons · *Objection correct, conclusion unaffected*

The race above has an asymmetry in it. The target is realised variance five trading days ahead; VIX prices a thirty-day horizon. A thirty-day option-implied variance is not the natural forecast of a five-day realised variance, so the section was run with a horse suited to a different distance. Cboe publishes VIX9D, a nine-day index on the same underlying computed the same way, from January 2011. Nine calendar days is about six trading days against the target's five, so this is near-matched rather than exactly matched; it is the closest listed instrument that exists, and its horizon is about a third of VIX's. Both blocks are given the same number of regressors, one implied-volatility series each plus VDAX-NEW on both sides, so the estimation cost of Section 6.1 is identical on the two sides and cancels in the difference.

The cost of the test belongs before its result. VIX9D begins in 2011, the decision variable of equation (2) needs a 252-day trailing median before it exists, and the walk-forward needs 1,500 days of labelled history before it can fit, which leaves 2,098 test days from May 2018, 46% of Table S22's window. It contains the 2020 episode but not the 2008 one, and 2008 is where being late costs a forecaster most. That sample is also simply harder: the identical thirty-day block loses between 0.0197 and 0.0274 of R^2 when it is moved to the shorter window with nothing else changed. We measure that separately so it cannot be mistaken for the horizon effect, and every column below sits inside the same window.

Table S19. The horse race with the horizons matched, 2,098 days, May 2018 to September 2026. IV30 is the block of Table S22, VIX at $t-1$ with VDAX-NEW; IV9 replaces VIX with VIX9D at the same regressor count; TERM carries both tenors. The fifth column is the horizon effect on a fixed sample and the last is what the seven foreign closes add *given* the horizon-matched block. Giacomini-White throughout: IV9 against IV30 is a comparison of two non-nested models of equal width, and the foreign increment is nested.

δ	domestic	+ IV30	+ IV9	+ TERM	IV9 − IV30	GW z	foreign | IV9	GW z
0	0.4861	0.5293	0.5528	0.5497	+0.0235	3.28	−0.0136	−2.21
5	0.3168	0.5050	0.5442	0.5409	+0.0392	3.08	−0.0174	−3.31
13	0.1920	0.5024	0.5424	0.5367	+0.0399	3.08	−0.0157	−3.31
21	0.1166	0.5036	0.5423	0.5364	+0.0387	3.08	−0.0150	−2.87
55	−0.0305	0.5064	0.5449	0.5429	+0.0385	2.90	−0.0135	−2.28

Two things are in that table and they point in opposite directions. The horizon objection is correct: matching it is worth between 0.0235 and 0.0399 of R^2, at every one of the five delays, with Giacomini-White statistics of 2.90 to 3.28. This is not a marginal improvement that survives on a technicality; the nine-day index is simply the better instrument for a five-day target, and Section 7 had been using the worse one. But the conclusion the section draws does not move. The last two columns are what the foreign cross-section adds on top of the horizon-matched block, and they are negative at every delay, with statistics between −2.21 and −3.31. The paper had been reporting redundancy against an implied-volatility control that was weaker than it should have been, and redundancy holds against

the stronger one. The horizon objection is right about the instrument and gets a firmer version of the same answer.

Two details close off the obvious follow-ups. Replacing the nine-day level with the nine-to-thirty-day slope, at the same regressor count, is worse at every delay, with statistics from −2.34 to −3.35: what the near end of the term structure carries is its level, not the shape of the curve. And carrying both tenors is slightly worse than carrying the nine-day one alone, 0.5497 against 0.5528 at zero delay, because the second series does not pay for its own coefficient, which is Section 6.1's mechanism again on a two-column block. One incidental is worth recording because it was not designed: on this window, which shares barely half its days with Table 2's, the foreign block's substitution rate comes out 51%, 64%, 68% and 73% at five, thirteen, twenty-one and fifty-five days, against 51%, 64%, 67% and 71% in Table 2. $R(\delta)$ barely moves between two half-overlapping samples measured eight years apart.

One objection to all of this has to be answered rather than noted, and it is the obvious one. This window contains exactly one violent episode, and a front-tenor index separating from a back one is precisely what a crisis in the term structure would produce, so the whole of Table S19 could be a fact about March 2020. The test is to delete it: every test day from 15 February to 30 June 2020 is removed from the evaluation, the models untouched, leaving 2,004 days. The nine-day index still wins at all five delays, by 0.0245 to 0.0405 of R^2 with statistics of 3.09 to 3.46, larger at four of the five delays than with 2020 included rather than smaller, and the foreign block still adds nothing, between −0.0147 and −0.0198 with statistics from −1.93 to −3.17. Neither conclusion in this subsection rests on the crisis.

S25. The two variance-scale losses, and why only one of them is Patton's · *Two losses distinguished, conclusion unaffected*

One distinction belongs here rather than in a footnote, because getting it wrong flatters the result and the two losses are easily confused. The wrong one reconstructs nothing: it computes squared error on the median-normalised variance, $\exp(y_t)$ against $\gamma_t \exp(f_t)$, which is Patton's MSE weighted by $1 / M^2$ whenever both sides are reconstructed with the same median. The weight is largest exactly where the median is smallest, so it is not a neutral rescaling, and the two losses are not the same loss: the substitution rate under the normalised version runs 6 to 29 points above the natural-scale one, by 10 points at eleven weeks. A forecaster cannot reconstruct both sides with the same median, because the outcome carries M_{t+h} and only M_t is known at the origin, so that gap is the weight plus the median's drift over h days rather than the weight alone. The table below splits it with a third, infeasible column and finds the drift worth at most 1.1 points at any delay, which leaves the weight carrying essentially all of the correction.

Mean absolute error, used in some recent work in this area, is in no version of the family. One choice inside that arrangement needs defending, because it looks backwards: the headline number in this paper is the one loss of the three that is NOT proxy-robust. We lead with log-scale R^2 because the paper's object is a ratio, $R(\delta)$, and a ratio needs a denominator a reader can hold: the share of a delay-induced loss recovered, naturally bounded near 0 and 100%, on the same scale as the companion note (Vellanikaran 2026) and as the decision variable itself. QLIKE skill on a variance scale is the right loss and the wrong ruler for that purpose; its levels are not comparable to anything else in this literature. The defence is that the choice is presentational and is made where it can be checked: both proxy-robust losses are reported beside the headline in every table where a conclusion turns on one, and

they support the same conclusions everywhere including the $\delta = 0$ row, where both point in the same negative direction but the natural-scale MSE is too imprecise to establish an effect. Because the magnitude there is about one point of R^2 and the practical question concerns stale rather than current marks, no economic claim in this paper rests on that row, and Section S9 says so. A reader who prefers the robust family throughout can read the QLIKE columns and will reach the same conclusions with different numbers attached.

Section 4.4 reports squared error on the natural variance scale and names the median-normalised computation as the thing it is not. This section gives both columns, because the size of the difference is what decides whether the distinction matters, and it is a number rather than a remark.

Two things separate the two columns, and only the first of them is a factor. The target is $y_t = \log(\sigma^2_{YZ,t+h} / M_{t+h})$, so $\exp(y_t)$ is a variance divided by a median and $\exp(y_t)\, M_{t+h}$ is the variance itself. Had the forecast been reconstructed with that same median, squared error on the normalised pair would be squared error on the natural pair weighted by exactly $1 / M^2_{t+h}$, and nothing else would differ. But M_{t+h} is not known at t. The only reconstruction a forecaster can run multiplies the forecast by M_t, and the natural-scale column reported throughout this paper is that one, so moving from the normalised column to it takes the weight off *and* moves the forecast's median h days at the same time. The gap between those two columns is therefore not, on its own, a measurement of the weight, which is the reading it invites.

Table S20 separates them with a third column nobody could compute. Reconstruct both sides with M_{t+h}: per day that loss is exactly M^2_{t+h} times the normalised loss, which the script checks numerically rather than asserting, to 1.50e−12 in relative terms. So what separates that column from the normalised one is the weight alone, and what separates it from the implementable one is the median's h-day drift alone. The weight turns out to carry nearly all of the correction, 28.4 of the 29.0 points at one day and 9.3 of the 10.3 at eleven weeks, with the drift never worth more than 1.1 points at any delay, so the earlier wording had the economics right and the algebra wrong, and what is on the record now is two named effects rather than one undifferentiated number.

The weight is not innocuous: it is largest in the calmest windows, which is exactly where the delay does least damage and the substitution rate's denominator is smallest. Patton's result is about the unweighted loss (2011). QLIKE is unaffected by any of this, and not by luck: it depends on the ratio of outcome to forecast, so a common factor cancels exactly, which Part 0 of the script demonstrates numerically to ten decimal places rather than asserting.

Table S20. The substitution rate under three variance-scale losses, and the correction decomposed. 'Normalised' divides outcome and forecast by the trailing median and is the column most easily mistaken for Patton's MSE. 'Both at t+h' reconstructs outcome and forecast with the same median M_{t+h}, which no forecaster could do, and differs from the normalised column by the $1/M^2$ weight alone. 'Natural' forecasts with the median known at the origin and is the implementable loss Patton's robustness result covers; it differs from the middle column by the median's h-day drift alone. The last three columns are those two differences and their sum, in points of the rate; an entry of ± 0.0 is a signed difference below a tenth of a point. This table carries no interval column, and the paragraph after it says why: under the fixed-b inversion of Section 4.6 neither variance-scale rate is identified at most delays.

δ	R(δ) normalised	R(δ) both at t+h	R(δ) natural	weight	drift	total
1	60.1%	31.7%	31.1%	+28.4	+0.6	+29.0
3	74.7%	60.5%	60.5%	+14.2	+0.0	+14.2
5	85.6%	73.6%	73.7%	+12.0	−0.1	+11.9
8	87.9%	78.7%	78.8%	+9.2	−0.1	+9.0
13	83.6%	75.7%	75.1%	+7.9	+0.6	+8.5
21	83.5%	75.4%	74.3%	+8.1	+1.1	+9.2
34	82.0%	76.1%	75.1%	+6.0	+0.9	+6.9
55	90.1%	80.8%	79.7%	+9.3	+1.1	+10.3

No interval is quoted for any column of this table, and the reason is a result rather than an omission. The rate is a ratio of two means under any of these losses, so the inversion of Section 4.6 applies to all three columns exactly as it applies to Table 2. Run on them, it declines to identify them. The normalised rate returns an *exclusion* set at all eight delays: two half-lines whose excluded middle begins no lower than 90 at any delay, so that at eleven weeks the set is everything outside [108, 182], and the natural-scale rate returns the whole real line at six of the eight delays and sets 219 and 170 points wide at the other two, 75% [39, 258] at $\delta = 34$ and 80% [34, 204] at $\delta = 55$. Across the four losses lab10 inverts, fourteen of thirty-two sets are not ordinary intervals, and every one of the fourteen is in these two columns.

That is what weak identification looks like when the procedure is allowed to report it, and it is the strongest defence this appendix can give for a choice this section defends above on presentational grounds. The denominator of the natural-scale rate is what a delay costs a forecaster measured in squared error on untransformed variance, a quantity dominated by a handful of crisis days, and a ratio whose denominator is estimated that imprecisely cannot be pinned down however it is bracketed. A percentile bootstrap of the same differentials returns finite endpoints at every one of these cells and so conceals that, which is why none of them is quoted here. The log-scale and QLIKE columns of Table 2 are identified at every delay, with denominator *t*-statistics of 5.3 to 6.4, and they are the columns every conclusion in this paper rests on.

The normalised loss overstated the rate at every delay, by 29 points at one day and by 10 at eleven weeks. Three things follow and the middle one is the reason this correction improved the paper rather than damaging it. The headline is unaffected, because the headline is the log-scale figure and neither variance-scale column feeds it. The two robust losses now agree: QLIKE reads 80% at eleven weeks and the natural-scale MSE reads 79.7%, where before the mislabelled column read 90% and sat

apart from both. And the $\delta = 0$ sign flip this appendix used to report was the artefact rather than the finding, which Section S9 now records.

The normalised column is kept here rather than deleted for the same reason the other corrections in this appendix are kept. A reader who finds an error in a paper wants to know its size, and a paper that quietly replaced one loss with another would be asking to be trusted on precisely the point it had just got wrong.

S26. A model of the delay curve

This section states a specification before the measurements that support it, because written down it turns the delay curve from a description into a prediction with a testable ceiling, and that ceiling is what Sections 5 to 7 are then measured against. The specification is not assumed: it is the shape two of those measurements take, and a reader who prefers the evidence first can read Sections 5 and 6 and return here.

Section S27 finds the recovery entirely in the leading component of the foreign block, with the six orthogonal components worth less than nothing. Section S4 finds that the level of volatility, which equation (2) removes, carries information that grows with delay. Together they say: log variance is a slow level, plus a factor several markets share, plus something idiosyncratic. For market i,

$$\log \sigma^2_{i,t} = \mu_{i,t} + \lambda_i \, g_t + e_{i,t}, \quad g_t = \varphi \, g_{t-1} + u_t, \tag{14}$$

with μ the slow level the 252-day median tracks, g a persistent global factor, e idiosyncratic and u white noise. The decision variable of equation (2) subtracts an estimate of μ, so $a_{i,t} \approx \lambda_i \, g_t + e_{i,t}$. That is the whole of Section S4 in one line: the normalisation removes μ by construction, so restoring it is worth something.

A forecaster δ days late sees $a_{SPX,t-\delta}$, one noisy reading of $g_{t-\delta}$. The cross-section supplies seven readings of g_t, whose common component suppresses the idiosyncratic part. Under (14),

$$E[\, g_{t+h} \mid g_{t-\delta} \,] = \varphi^{\,h+\delta} \, g_{t-\delta}, \quad E[\, g_{t+h} \mid g_t \,] = \varphi^{\,h} \, g_t, \tag{15}$$

and three consequences follow. Two are checkable against quantities measured elsewhere in this paper and not fitted to the curve; the third, below, turns out to be algebraically tied to its own endpoint and is reported as a reading rather than as a test.

First, the stale forecaster's skill decays geometrically: $\log S_{own}(\delta)$ is linear in δ with slope $2 \log \varphi$. Fitted over the eight delays from one day to seven weeks, the line has $R^2 = 0.9980$ and slope -0.05424 per day, implying $\varphi = 0.9732$. The factor's own persistence, estimated from the principal-component series and never shown the decay curve, is 0.9678. The two differ by 0.0055. Two delays are held out and the reason belongs here rather than in the code, because one of them is the most influential point in the column. The derivative of a log is $1/S$, so at $S_{own}(55) = 0.0072$ a three-thousandth move in skill shifts $\log S_{own}$ by 0.42 against 0.006 at zero delay: in an unweighted least squares on logs that point carries seventy times the leverage of the first one. Zero delay is held out as the intercept the line is measured against. Both readings are reported, because excluding the most influential point is the move a reader should distrust: on all ten delays the line fits worse, $R^2 = 0.9692$ against 0.9980, and lands *closer* to the factor's persistence, $\varphi = 0.9647$ against a measured 0.9678, a difference of 0.0031. So the held-out points cost the linearity claim and do not cost the agreement the section rests on, and the

linearity claim is the weaker of the two on this evidence. The delay curve is the factor's memory, seen through a forecast.

Second, $S_{cross}(\delta)$ should be near-flat, because the foreign block delivers g_t whatever the domestic delay is. Across eleven weeks it falls 27%, against 99% for the domestic model. Perfect flatness is not predicted, because the domestic block is still in the cross-sectional model and still decays, but the contrast is the model's.

Third, and most useful: $R(\delta)$ rises from zero towards a ceiling below one, and the model names it. Since $S_{own}(\delta) \to 0$,

$$R(\infty) = S_{cross} \div S_{own}(0), \quad (16)$$

which is below one here, and the reason is worth stating as the inequality it is rather than the implication it looks like. Two forces act against each other: the peers never observe $e_{SPX,t}$, which costs them to the extent that e_{SPX} persists, while averaging seven of them measures g_t *more cleanly* than the target's own single contaminated value does. A ceiling below one needs the first to dominate, which is a statement about how persistent the target's idiosyncratic component is relative to the factor, not about whether it persists at all. Simulate the specification with the idiosyncratic component made less persistent than the factor and the ceiling crosses one, with the cross-section beating a fresh own mark outright. No figure is quoted for that here because this appendix quotes only figures its own scripts produce, and it is not a claim this panel tests. What the panel settles is the direction: the measured ceiling is below one, and that is how this section uses equation (16). Both quantities are measured elsewhere: $S_{cross}(55) = 0.3729$ and $S_{own}(0) = 0.5188$, giving a ceiling of 71.9%, against a measured 69.7% at $\delta = 34$ and 71.5% at $\delta = 55$.

That agreement is weaker evidence than it looks, and the reason is arithmetic rather than empirical. Write the rate out: $R(\delta) = [S_{cross}(\delta) - S_{own}(\delta)] \div [S_{own}(0) - S_{own}(\delta)]$. When $S_{own}(\delta)$ is near zero (at $\delta = 55$ it is 0.0072), this collapses to $S_{cross}(55) \div S_{own}(0)$, which is equation (16). So the "prediction" and the measurement are built from the same two numbers and must agree; the 0.4-point gap between 71.88% and 71.48% is the size of $S_{own}(55)$, not the accuracy of a forecast.

We therefore call (16) a factor-model *interpretation* of the observed plateau rather than a prediction of it. What the model earns is the reading: the curve flattens because the cross-section's skill is a factor share bounded by how well seven peers estimate g, and the level it flattens at is that share. What it does not earn is credit for having predicted a number it is algebraically tied to. A genuine out-of-sample test would estimate the ceiling from parameters independent of the endpoint being predicted (the factor's loadings and its persistence, say, rather than $S_{cross}(55)$ itself), and this paper has not done that.

So the substitution rate is not an arbitrary rising function. It is a factor share, it converges, and the paper can say what it converges to. The model also explains why Section S27's orthogonal remainder is worthless, being e, which forecasts nothing, and why Section S12's staleness penalty exists, since an aged foreign block reads g at a date earlier than t, which (15) prices at φ per day. Three results the paper had separately are one mechanism.

What the model does not explain is why the mechanism is stronger in stress. Equation (15) holds λ and φ constant; Section S5 shows the factor doing more work when markets are frightened, which

would need λ or the variance of u to move with state. We have not fitted that, and it is the obvious next specification.

S27. Breadth against its own leading component

A further objection is omitted-variable bias: foreign breadth may not be capturing cross-border diffusion at all, but proxying for a global systemic risk factor, so that what looks like transmission is simultaneous reaction to shared news. Section 6.1 is most of the way to agreeing, since one real-time factor beats all seven series at every delay, but it never separates the two halves and asks which one carries the recovery. The decomposition is available at no cost, because the basis is already being estimated.

At each refit, write v_1 for the leading eigenvector of the training covariance of the standardised foreign block. Let p_t be that block's seven decision variables at t, standardised on the training window, and let v_1 be the unit eigenvector belonging to the largest eigenvalue of their training covariance, with its sign pinned so that the first loading is positive. The split is

$$p_t = g_t\, v_1 + r_t, \quad g_t = p_t' v_1, \quad r_t = p_t - g_t\, v_1, \tag{17}$$

where g_t is the scalar global factor and r_t is the six-dimensional remainder orthogonal to v_1. Sign pinning is not cosmetic: an eigenvector is defined only up to sign, and one that flipped between refits would enter the regression as noise unrelated to the data. The + *PC1* column below uses g_t as a single regressor and + *remainder* uses the six components of r_t. Each day's standardised row x splits into a global part $(x \cdot v_1)v_1$ and a remainder orthogonal to it, spanned by components two through seven. Adding the two back together is a rotation of the raw block and must reproduce it: it does, to within 0.0010 of R^2 at every delay, the check that licenses reading the split.

Table S21. The foreign block split into its global component and the remainder orthogonal to it, both estimated in real time at every refit. The last column tests PC1 against the remainder by Giacomini-White; positive favours the global component.

δ	domestic	+ PC1	+ remainder	+ all seven	PC1 − domestic	remainder − domestic	GW z
0	0.5188	0.5200	0.5080	0.5089	+0.0012	−0.0108	2.77
3	0.4094	0.4609	0.3988	0.4492	+0.0515	−0.0106	4.43
5	0.3637	0.4551	0.3545	0.4434	+0.0914	−0.0093	4.89
13	0.2332	0.4269	0.2263	0.4150	+0.1937	−0.0069	4.73
21	0.1478	0.4117	0.1344	0.3973	+0.2638	−0.0134	4.08
55	0.0072	0.3918	−0.0190	0.3729	+0.3846	−0.0262	4.29

The answer is consistent across the delays tested. The global component alone beats the domestic model at every delay, and significantly at five of the six: the exception is $\delta = 0$, where the increment is +0.0012. At eleven weeks it is worth 0.3846 of R^2. The remainder, six real-time components orthogonal to it, is *negative* at every delay: it beats the domestic model at none, and carrying it costs between 0.0069 and 0.0262 of R^2, the estimation bill of Section 6 with no information to pay for it. Giacomini-White separates the two at every delay, from z = 2.77 to 4.89.

So the objection is right about the mechanism and we adopt the correction. What this paper measures is a global volatility factor observed while the domestic series is stale, not bilateral diffusion between particular markets. Three things follow. The substitution rate is unchanged: it does not matter to the number what the factor is called. The practical advice sharpens: a forecaster wants a cross-market average, not a panel of countries, as Section 6.1 already recommended for a different reason. And a limitation appears that Section 10 now carries: a first principal component of seven equity indices is a proxy for global risk, not a measurement of it, so orthogonalising against it is weaker than orthogonalising against a factor built from a wider universe, and the remainder may still hold global content this basis cannot see.

S28. The implied-volatility race in full

Section 7 of the paper states this race's result and points here for the working. The race itself is below, followed by the three questions it raises.

The objection this section answers is the one Section 2 names as unavoidable: if a forecaster whose domestic data is δ days stale can see today's VIX, why would they need Tokyo? Implied volatility is forward-looking, it never goes stale, and both Korkusuz (2025) and Buncic and Gisler (2016) report that much of what looks like a cross-market effect is attributable to it. So the two candidates go into the same walk-forward, on the same labels, with the same estimator. Four models, all with the domestic block held δ days stale: domestic alone, plus the foreign block, plus implied volatility, and plus both. The number the objection turns on is what the foreign block adds *given* implied volatility. VIX enters at $t-1$ throughout, the strictly admissible specification of Section 4; the same-day arm is reported below as a sensitivity and reaches the same verdict. VDAX-NEW begins in May 2001 against the index panel's February 2000, so all four are re-estimated on the same reduced window of 4,525 days and the comparison stays internal.

Table S22. Out-of-sample R^2 for four information sets, 4,525 days, September 2008 to September 2026. The last two columns are what the foreign block adds *given* implied volatility, and the Giacomini-White statistic testing it. VIX enters at $t-1$, the strictly admissible specification of Section 4; the same-day timing sensitivity check follows in the text.

δ	domestic	+ foreign	+ implied vol	+ both	foreign | IV	GW z
0	0.5138	0.5048	0.5545	0.5442	−0.0103	−2.27
3	0.4042	0.4475	0.5334	0.5229	−0.0105	−1.96
5	0.3587	0.4427	0.5324	0.5229	−0.0095	−1.67
13	0.2276	0.4142	0.5258	0.5167	−0.0090	−1.35
21	0.1401	0.3969	0.5243	0.5146	−0.0097	−1.42
55	0.0009	0.3761	0.5261	0.5166	−0.0095	−1.30

The third column is the one that decides the section. A forecaster holding implied volatility loses almost nothing to delay: R^2 goes from 0.5545 to 0.5261 across eleven weeks, where the domestic-only forecaster falls from 0.5138 to 0.0009. Put the other way, a desk with VIX and VDAX and a three-week-old mark still beats a desk with fully current domestic data and no options information, 0.5243 against 0.5138. The last two columns are the comparison the objection turns on, and they are negative at every delay: adding seven foreign closes on top of implied volatility makes the forecast worse, not better, by about one point of R^2. That reads like harm and the subsection below shows it is a bill, the

cost of estimating seven coefficients, but on the question the section asks, the answer is already here. Breadth adds nothing measurable to a forecaster who can see the options market.



Figure S1. The dotted green curve is almost flat. A forecaster holding implied volatility barely notices how old their domestic data is, and from about four days on they are ahead of where the domestic-only forecaster started. The long-dashed curve, which adds foreign closes on top, sits just below it at every delay.

The dotted green curve is the result. Implied volatility does not merely slow the decay, it very nearly stops it: R^2 falls from 0.5545 to 0.5261 across eleven weeks, against 0.5138 to +0.0009 for the domestic model alone. Expressed as a substitution rate that is 103% at three weeks, and a rate above 100% is the signal from Section 4 that the measure has changed meaning. A forecaster holding three-week-old domestic data and today's VIX scores 0.5243, above the 0.5138 of a forecaster with fully current domestic data and no options market. Implied volatility is not restoring what staleness took; it is supplying something the realised-variance history never contained.

That should not be mysterious, though it should not be overstated either. The VIX is a RISK-NEUTRAL expectation of S&P variance over roughly the coming MONTH; our target is REALISED variance over the coming five days. The two differ in horizon, by a factor of about four, and in measure, by a variance risk premium that is itself time-varying. So the VIX is not the quantity we are predicting. It is, however, a purpose-built forward-looking forecast of something close to it, and comparing that with a cross-section of correlated state variables is comparing a forecast with a set of clues. Read that way the surprise is not that implied volatility wins but that seven foreign equity markets get within two-thirds of it despite the horizon and measure mismatch working against the options market.

The sixth column is what the objection actually turns on, and it is negative at every delay. Given implied volatility, adding the foreign block makes the forecast slightly worse, and the Giacomini-White statistic, for which the domestic-plus-IV model is nested inside the full one so the same test applies as in Section S2, reaches conventional significance at the short delays. Taken literally this says foreign closes damage a forecast that already has VIX, which would be strange behaviour for genuine information.

Section 6 supplies the alternative reading, and the same instrument settles it. Replacing the seven foreign series with AR(1) surrogates matched on persistence and variance leaves regressors that

carry nothing by construction, and what remains is the bill for estimating them. It comes to roughly 0.012 to 0.018 of R^2, the whole of the negative. The content of the foreign block given implied volatility, gross of that cost, is +0.002 at $\delta = 0$, +0.006 at $\delta = 5$, +0.008 at $\delta = 21$ and +0.008 at $\delta = 55$, and bootstrapping each of those the way Section 6 describes gives intervals that all cover zero: $[-0.007, +0.012]$, $[-0.007, +0.018]$, $[-0.006, +0.021]$ and $[-0.007, +0.022]$. Those four are measured on the admissible arm of Table S22, the one with VIX lagged a full day; the decomposition used to be run on the same-day arm alone, which is the specification this paper declines, and its bound was narrower. The correct word is redundant, and it is worth saying exactly what that word is doing. It records a failure to distinguish, not a demonstration of equality: the widest of the four intervals above rules out a gross gain larger than 0.022 of R^2, at $\delta = 55$ and leaves anything smaller open, so what is established is a bound and not a zero. Within that bound, foreign closes do not damage a forecast that already sees the options market; they add nothing we can detect, and paying to estimate nothing is what makes the measured figure negative.

The table above is the admissible specification: VIX lagged a full day, which discards a day of information the live index would have carried. Substituting the same-day close, which contains fifteen minutes we are not entitled to, moves the options market the other way: R^2 at $\delta = 21$ rises from 0.5243 to 0.5837, a substitution rate of 119%, while the incremental contribution of the foreign block stays negative at all ten delays and its gross content stays indistinguishable from zero. Neither arm is the specification we would choose, and the one we would choose is not available to us. What licenses a conclusion here is not that the 21:00 observation falls between these two in time, which would say nothing about how it forecasts, but that the verdict is the same under a specification that gives the options market a full day less information and under one that gives it fifteen minutes more. A finding that survives that much variation in the input is not resting on the timing choice.

One feature of that result deserves a sentence. VIX is a thirty-calendar-day implied volatility and the target here is realised variance over five trading days, and the two are not in the same unit: thirty calendar days is about twenty-one trading days, so the winning instrument is measured over a horizon about four times the length of the thing it is forecasting, not six. That it dominates anyway says something about how much of the five-day question is answered by the level of the whole term structure, not by its front. The matched instrument exists, since Cboe publishes a nine-day index, VIX9D, formerly VXST, and Section S24 runs the race again with it. The horizon does matter, by rather more than we expected, and the conclusion of this section does not.

Three further questions about this race are answered below: whether redundancy survives compressing the cross-section, what the skill column means economically, and three ways the race could have been unfair to breadth.

S28.1 One regressor instead of seven

The test above adds seven regressors to the implied-volatility model, and it therefore cannot separate two different reasons for a null: the foreign block carries nothing beyond VIX and VDAX, or it carries something and the cost of estimating seven coefficients consumes it. Section 6 says that cost is real and Section 6.1 says a compression avoids part of it, so the confound is not hypothetical. The compression supplies the fix: add *one* regressor instead of seven.

Table S23. What a single compressed summary of the foreign block adds on top of stale domestic data and the implied-volatility block, which is VIX and VDAX-NEW. The mean requires no estimation at all; PC1 is the real-time first component of Section 6.1. The seven-regressor column is Table S22's, for comparison. Both arms are among the nine specifications Section S6 prices with a Reality Check and an SPA test; this grid is not one of the secondary grids counted in the Section S37 pool, which covers the five families listed there. Cells significant only at the uncorrected 5% level are *exploratory* and no claim in this paper rests on one of them alone.

δ	+ IV only	+ mean	+ PC1	+ all seven	mean | IV	GW z	PC1 | IV	GW z
0	0.5545	0.5557	0.5556	0.5442	+0.0012	0.43	+0.0012	0.42
3	0.5334	0.5356	0.5358	0.5229	+0.0021	0.84	+0.0024	0.94
5	0.5324	0.5351	0.5354	0.5229	+0.0027	0.85	+0.0030	0.94
13	0.5258	0.5292	0.5297	0.5167	+0.0035	0.86	+0.0040	0.97
21	0.5243	0.5279	0.5282	0.5146	+0.0036	0.87	+0.0040	0.95
55	0.5261	0.5300	0.5301	0.5166	+0.0039	1.04	+0.0040	1.03

The sign flips and the conclusion does not. With seven regressors the increment is negative at every delay; with one it is positive at every delay. That is the estimation-cost mechanism of Section 6 showing up exactly where it was predicted to, and it means the negative in Table S22 was a bill rather than evidence of harm, a reading the paper offered there and can now demonstrate rather than argue. But no Giacomini-White statistic reaches 1.96: the largest is 1.04, at eleven weeks. A single cheap summary of seven foreign markets, carrying essentially no estimation cost, still adds nothing detectable to a forecaster holding implied volatility.

That is a stronger statement than the one this section made before, and it should be read as strictly stronger, a sharpening of the earlier finding. The earlier test could be dismissed as penalising the foreign block for its parameter count. This one cannot. What it cannot do either is prove absence: increments of +0.001 to +0.004 with statistics between 0.42 and 1.04 are consistent with a small positive contribution this sample is too small to resolve, and the honest summary is that the foreign block adds nothing we can detect given implied volatility, not that it adds nothing.

S28.2 Economic interpretation of the skill column

A point of R^2 on log variance is not an intuitive quantity, and the reading a risk manager needs is narrower than the column suggests. What breadth repairs is a stale forecaster's tendency to over-estimate variance; the understating tail is unmoved: it runs between 8.6% and 11.5% of days whatever the delay and whichever model is used. Concretely, at eleven weeks the stale domestic forecaster's variance forecast is off by a median factor of 1.90, and is more than a factor of two out on 59.2% of days against 37.1% for the forecaster holding the cross-section, against 31.1% for a forecaster with no delay at all. Breadth closes most of that gap, and all of what it closes is over-estimation. Anyone reaching for the cross-section as protection against being surprised by a volatility spike is reaching for the wrong instrument.

S28.3 Three ways the race could be unfair

Three objections to the comparison above can each be tested rather than argued about, and all three are tested below so that this section stays the length of its result. None of them overturns it.

Redundancy was established with a ridge, and a linear map might simply be too blunt to find what the foreign block holds given implied volatility. Section S22 gives the block two richer maps: a random Fourier feature expansion at two widths, which is a one-hidden-layer network whose first layer is drawn rather than learned, and a k-nearest-neighbour arm imposing no functional form at all. Nothing appears. Every statistic that would have overturned the redundancy result comes back negative, and the richer maps cost more than the plain block does.

Every number above forecasts five trading days ahead, and Section S23 re-runs the whole experiment at one day, two weeks and one month to establish that this is a convention rather than a finding. The rate is stable across the middle of that range and falls at the ends for reasons the section separates: at one day the target is an object the forecaster can largely already see, and at one month it has dropped to 50% with an interval too wide to lean on. Section 10 treats that as a limit on how far the result reaches rather than as a result.

One asymmetry in the race is worth removing rather than defending. The target is realised variance five days ahead and VIX prices a thirty-day horizon, so the section above ran the options arm with a horse suited to a different distance. Cboe's VIX9D is the closest listed instrument, with a horizon about a third of VIX's and near-matched to the target rather than exactly matched, and Section S24 runs the race again with it, giving both blocks the same number of regressors so that the estimation cost of Section 6.1 cancels in the difference. The horizon objection is correct and matching it is worth something measurable; the conclusion of Section 7 survives it, and the section states plainly what the shorter sample the test requires costs.

S29. Feasibility at the asset's reporting frequency

The larger absence is the one Sections 8 and S30 keep naming, and this paper has treated it as though the difficulty were finding an appraisal-based series. That is not the difficulty, and being precise about what is changes what a future study should do. The proxy is Yang-Zhang over five daily bars, which needs an open, a high, a low and a close for every session, and an appraised asset has none of those: it has one number per quarter. The walk-forward then needs 252 + 1,250 + 250 = 1,752 observations before its first forecast, which at one mark per quarter is four hundred and thirty-eight years. No dataset solves that. The design has to be re-specified at the frequency the asset reports, and the question nobody had asked is whether the measurement survives when it is.

It does. Coarsening the S&P and its seven peers to month-end marks throws away exactly what an appraised series never had, leaving the proxy to be built from the variance of the last six monthly returns against a sixty-month median, with the delay counted in months and every window scaled by twenty-one. On 109 test months the rate at a one-quarter delay is 51.9% [37, 79], and at six months 57.5% [48, 79]. Smoothing the mark by the partial adjustment of Section S15, which is how the property literature has described appraisal indices for thirty years, leaves 45.5% [31, 76] and 55.8% [46, 80]. All four are quoted to the decimal Table S24 carries. The appraised quarterly figure is the reason: the lab prints it as 45.5% in the table and as 45% in its own verdict, so its value lies below the half, and rounding the rounded 45.5% a second time had put 46% in this sentence.

Table S24. The same experiment on month-end marks alone, with no daily bars anywhere in it. The delay is in months, so three months is one quarterly marking cycle. The appraised column applies the partial adjustment of Section S15 to the target before anything else is computed. Intervals are moving-block bootstraps at twelve months.

delay	R^2 own	R^2 cross	R, clean marks	R, appraised marks
0	0.7694	0.7359	n/a	n/a
1 month	0.4846	0.5555	24.9% [−14, 65]	24.9% [−11, 72]
2 months	0.2331	0.4540	41.2% [14, 71]	37.1% [16, 75]
3 months	0.0725	0.4340	51.9% [37, 79]	45.5% [31, 76]
6 months	−0.1399	0.3832	57.5% [48, 79]	55.8% [46, 80]

Two things in that table are worth more than the point estimates. The shape is the daily result: at one month the rate cannot be told from zero, exactly as it cannot at one day in Table 2, and it climbs from there towards a ceiling. And the appraised column is barely different from the clean one, which is Section S15's equivalence holding at a frequency where it was never tested. The apparatus is not daily by nature.

What the intervals will not do is price the effect. Forty-one to forty-six points of width at the quarterly marking cycle establishes that it is there and cannot say what it is worth. That gap closes with history rather than with method, and the arithmetic is worth stating because it tells a future study what to look for. A bootstrap interval narrows roughly with the square root of the test window, so reaching a width of forty points from the forty-one observed here needs about 115 test months, and a width of thirty needs about 205, which with the burn-in comes to twenty-seven and thirty-four years of marks. That calculation assumes independent draws and so understates what a persistent series requires; it is a floor. It is also within reach: appraisal-based property indices and quarterly private-equity series with three or four decades of history exist, and the cross-section they would be matched against is public and free. So what was missing was a series and not a method, and Section 8 supplies the series.

S30. What a practitioner should do with this

Someone holding a stale mark on an asset with a liquid options market should prefer own-asset implied volatility for this forecasting objective, at the tenor matching their forecast horizon rather than at the headline one. Section 7 provides strong evidence for the first, and the foreign cross-section adds nothing measurable once implied volatility is in hand. Section S28 provides it for the second: for a five-day question the nine-day index beats thirty-day VIX at every delay we test, by 0.0235 to 0.0399 of R^2, which is a larger prize than most of what the rest of this paper argues over and costs nothing but choosing the right series.

Someone holding a stale mark on an asset with no options market has breadth as the only candidate on the table, and it is no corner case: it is the case Section 1 opens with. Section S13 puts a measurement behind that sentence for the first time. Where the only options market a forecaster can see is written on another index, implied volatility and breadth come out three points apart rather than thirty-one, and a compressed cross-section adds significantly in thirteen of twenty-four cells against none of eight where the option chain is on the asset itself. Buying someone else's options market is not the substitute that buying your own is. Quarterly-marked private equity, appraised property and funds reporting a lagged net asset value have no listed option chain on the asset itself,

so the substitute that wins Section 7 is not available to any of them. For appraised property the number no longer has to be extrapolated from equities at all: Section 8 measures it on twenty metropolitan house price indices and finds 67.9% [43, 91] of the loss from a quarter's extra staleness recovered on the return and 18.8% [6, 31] on the variance, with the peer cross-section compressed to a single column, which is the same recommendation the rest of this section makes for a different reason. The redundancy result therefore narrows where the equity measurement transfers, not where the question matters. Before buying anything, though, there is a cheaper thing to do, and Section S14 is what makes it worth saying. Correlate the stale series with the public cross-section over the days already in hand. Across six markets that one number tracks the substitution rate at +0.92, it runs from 0.612 to 0.808 across targets that all look like ordinary equity indices, and where it is small the cross-section repaired nothing and charged for the attempt. A holder whose asset is weakly coupled to public markets should expect breadth to do nothing for them, and can learn that before they buy a feed rather than after. Sections 4 and 5 show how that worth can be measured and give a liquid-market benchmark for it, rather than an estimate for the illiquid asset itself: about 80% of the delay-induced loss recovered in stressed markets, where the interval is the narrowest of the three states, 81% at eleven weeks with an interval of [67, 90], and an amount this sample cannot determine in calm ones. One practical point belongs with those numbers and is easy to miss, because the paper reports it as a robustness result. The rates above come from the original seven-regressor specification, which is not the model to run. Section 6.1 shows a single real-time factor beating all seven series at every delay and raising $R(55)$ to 75.2%, and an equal-weighted mean of the seven does nearly as well for no estimation at all. The seven coefficients cost one to five points of R^2 to fit and buy nothing back. Anyone building this should compress the cross-section first and treat the seven-regressor figures as the conservative benchmark they are. The unconditional figure is 71% against the paper's own control and 67% against the stronger control of Section S4; we quote the conditional numbers because they are the ones that mean anything. Against that sits a cost of one to five points of R^2 for carrying the extra series. One condition comes with those figures, and Section S12 makes it explicit: the foreign data has to be current. Every rate quoted in this paper assumes the cross-section arrives the same evening, and a feed one day behind turns Table S8's 36% into 9% at three days and its 51% into 29% at five. Before buying breadth, check that it will arrive on time. A cross-section older than the mark it is replacing is worth nothing, and at the shortest delays it is worth less than nothing.

One generalisation is worth refusing explicitly. It would be convenient to present Figure 2 as a benchmark for how quickly information decays in delayed systems, and it is not one. Those curves are a measurement of one index against seven specific peers over one sample. The correlation between a private portfolio and the public cross-section is a different quantity, and nothing here bounds it. A reader who wants a number for their own asset will not find it in this paper.

What does carry across is the apparatus rather than the numbers. $R(\delta)$ is defined in units that survive a change of loss function, market or candidate. The delay sweep is a design anyone with a delayed series and a candidate substitute can run, and the horse race in Section 7 is the question to ask first, because its answer decides whether breadth is worth buying at all. The cost decomposition in Section 6 separates what a source knows from what a forecaster keeps, the distinction that turned our own most confusing result into an ordinary one. Pointed at an appraisal-based or quarterly-marked series, that machinery would produce the numbers this paper cannot, and the check above says the delay model it carries is the right shape for that series, even though the numbers are not.

S31. The house price measurement in full

Section 8 of the paper states what this measurement settles and points here for the working. The measurement is below, followed by the three supporting results it rests on.

Before the measurement itself, Section S29 establishes that the experiment is feasible at an illiquid asset's own reporting frequency, on month-end marks with no daily bars anywhere in it.

S31.1 Twenty metropolitan house price indices

Three supporting measurements are held back to the end of this section: the panel's properties measured before any rate was estimated, the full pricing of the three candidate peer blocks, and the seasonal placebo.

Two things change from Section 4 and both are worth stating plainly, because a result that moved two variables at once and reported one of them would not be worth much. The asset class changes. The target variable also changes, because a monthly index has no intraday range and Yang-Zhang cannot be computed on it at all. The two are therefore carried separately throughout. One target is next month's log return, which is the quantity an owner of the asset cares about and is not this paper's target. The other is the log of trailing twelve-month realised variance over its own five-year median. That second target needs a caveat rather than a claim of equivalence. This paper's own target is forward-looking: realised variance over the five days *after* the forecast origin, none of which the forecaster has seen. The housing one is backward-looking, and forecasting next month's value of it means forecasting a quantity whose window overlaps the current one in eleven of twelve months. An own-only R^2 near 0.95 follows from that arithmetic and is not evidence that house price variance is easy to forecast. It also shrinks the denominator $R(\delta)$ divides by, which is why the rates on this coordinate are lower and why we do not read them as the same measurement made in a second asset class. The return target is the closer analogue of the paper's, and where the two coordinates are given different weight below, that is the reason.

Delta needs saying carefully too. All twenty metros carry the same publication lag of roughly two months, so delta is not that lag. It is what it is everywhere else in this paper: the amount by which the target's own mark is staler than the cross-section it sits in. The concrete case is an owner of property in one metro whose book is appraised on a cycle running behind the published index for the others. Three months is one quarterly appraisal cycle and is the row to read first.

The peer block has to be chosen before any rate is read, and at this sample size the choice is not cosmetic. Section S28 and Section S30 already recommend compressing the cross-section to a single real-time factor rather than fitting seven separate series, for the reason Section 6 prices. The bill is far larger here: the training window is 144 months rather than 1,250 days, and on the variance target the own-only model already reaches an R^2 of 0.9558, so thirteen peer columns have almost nothing left to earn and a great deal to spend. Table S27 prices three blocks as a set, the way Section S4 prices eight controls, and selects one by a rule fixed in advance: take the block with the highest cross-arm skill at zero delay, treat differences below 0.01 as immaterial, and break ties towards the smaller block. That statistic is measured where the own mark is fresh and there is nothing to substitute for, so it cannot favour a high rate, and the tie-break runs towards fewer regressors, which is the direction that lowers it.

Table S25. The substitution rate on fourteen metropolitan house price indices, under the peer block selected in Table S27. The delay is in months and is relative staleness, not the publication lag every metro shares: three months is one quarterly appraisal cycle. Eleven month-of-year dummies enter both arms. Intervals are moving-block bootstraps at twelve months, resampling blocks common to all fourteen metros so that one national housing cycle is not counted as fourteen independent ones. The last column counts metros whose own rate is positive.

delay	R^2 own	R^2 cross	R	metros with R > 0
target: next month's return; 311 test months per metro, 2000-08 to 2026-06				
0	0.7242	0.7404	n/a	n/a
1 month	0.5696	0.6580	152.7% [−78, 208]	14 of 14
2 months	0.3928	0.5965	66.2% [53, 86]	14 of 14
3 months	0.3506	0.5953	67.9% [43, 91]	14 of 14
6 months	0.2079	0.5731	69.1% [58, 79]	14 of 14
target: next month's trailing twelve-month variance (backward-looking); 251 test months per metro, 2005-08 to 2026-06				
0	0.9558	0.9543	n/a	n/a
1 month	0.8666	0.8693	0.6% [−10, 9]	6 of 14
2 months	0.7522	0.7742	10.2% [−2, 22]	10 of 14
3 months	0.6278	0.6852	18.8% [6, 31]	13 of 14
6 months	0.2695	0.4987	34.8% [18, 48]	14 of 14

Both coordinates exclude zero at a quarterly staleness. On the return target the rate is 67.9% [43, 91] and is positive on all fourteen metros. On the backward-looking variance coordinate it is 18.8% [6, 31] and positive on thirteen of fourteen, and that figure is the weaker of the two for the reason given above: most of next month's trailing variance is already in the forecaster's hands, so there is little for delay to destroy and correspondingly little for breadth to put back. The exception is Las Vegas, where the peer block does not merely fail to help but actively subtracts, taking 0.4232 down to 0.3631. Coupling does not account for it, though Las Vegas is the fourth lowest on the panel: Denver at 0.704, Charlotte at 0.699 and Cleveland at 0.661 all couple lower and all return positive rates, of 4.6%, 45.1% and 9.7%. So coupling does not pick this metro out, and no explanation is offered for it. It is one failure in fourteen on a panel whose coupling spread is too narrow to order anything, and it is recorded rather than accounted for. The shape of Table 2 survives as well: at one month the rate cannot be told from zero on either coordinate, exactly as it cannot at one day in Table 2, and it climbs from there. At one month on the return target the mean of the fourteen per-metro ratios reads 152.7%, with an interval of [−78, 208] that we read as the estimator declining to answer rather than as a rate above one. The explanation is the one Section S18 gives for preferring a pooled ratio on this panel: with a one-month delay there is almost nothing for staleness to destroy, so a single metro whose own delay cost is near zero dominates a mean of ratios. Pooling instead - dividing the average increment by the average delay cost, 0.6580 against 0.5696 over 0.7242 against 0.5696 - gives 57%, which is in line with the other delays and carries no suggestion of exceeding one. A fresh cross-section can beat the target's own fresh mark, which is true at zero delay on this panel, where the cross arm scores 0.7404 against 0.7242; that is not what produces this cell.

The first objection is the index's own construction. Each published figure for month t is a three-month moving average of sales pairs closing in months t − 2 to t, so consecutive monthly changes are built from overlapping transaction windows, and a rate manufactured by that smoother would look

exactly like a rate. Section S18 removes every shared transaction month by reading the index only every third month, on each of the three phases of the quarterly grid, and the artefact explanation predicts zero there. Pooled across metros the rate is 57% at one quarter of staleness, spanning 49% to 63% across phases, positive in 35 of 42 metro-phases, with each phase's interval excluding zero. So the explanation is not available; nor is the rate unchanged, since 57% is visibly below 67.9%, which is what one would expect if a modest part of the monthly figure is the smoother. De-smoothing the panel says the same thing from the other side and adds a finding: with the moving average inverted out, one month of staleness costs −0.0012 of R^2, nothing measurable and if anything the wrong sign, so no rate is quotable at all, while the peer block still adds +0.1351 of R^2 at three months on fourteen metros of fourteen. In this panel, smoothing accounts for most of what a short delay costs, which is consistent with breadth paying on assets that are marked this way. Whether that holds outside a repeat-sales index is not something fourteen metros can establish.

What this section settles, and what it does not. It settles that the central claim of this paper is not a fact about liquid equity indices. The claim holds on an asset class with no intraday range, no option chain and a mark that is smoothed by construction; it holds on the target variable the asset's owner cares about and on the one this paper was written about; it holds under a peer block chosen by a rule that could not see the rate; and it is not a calendar effect. What it does not do is price the effect more tightly than the intervals in Table S25, or test the slope of the coupling relation, for the reason given before the rates were estimated. And it remains an index study: nobody holds a Case-Shiller index, and a real appraised book would differ from one in ways this panel cannot show. What is no longer true is the sentence this paper carried until now, that the illiquid case rests on argument alone. It now rests on a measurement one step removed from the asset the argument is about.

S31.2 The panel, the peer blocks and the seasonal placebo

Section S29 established that the design survives being re-specified at monthly frequency, and closed by saying what was then missing: a series with two to four decades of marks and a public cross-section, run at the asset's own frequency. That series exists and it is free. The S&P Cotality Case-Shiller indices cover twenty United States metropolitan areas to July 2026, 9,163 published figures in all. Fourteen of them run monthly from January 1987; Minneapolis and Phoenix begin in 1989, Seattle in 1990, Atlanta and Detroit in 1991, and Dallas not until 2000. Rates are estimated on the fourteen with the full history, which is why a panel of twenty is reported as fourteen of fourteen, and the six later starters enter as peers wherever they exist. Part B below widens the peer block to all nineteen. These indices are the object this paper has spent nine sections arguing about rather than a proxy for it. Each monthly figure is a three-month moving average of closings and closings lag contract by weeks, so the mark is smoothed and stale before anyone touches it. No option chain is written on any of the twenty, so the implied-volatility substitute that dominates Section 7 is not merely weak here, it is absent. And all twenty publish on the same day, so when one is treated as stale, nineteen fresh peer marks exist: the construction of Section 4, transplanted whole.

Before any rate was estimated, two properties of the panel were measured, and both bear on claims made earlier in this paper. The first is persistence. Section S15 models a stale mark as partial adjustment at rho = 0.6, a figure taken from the property literature, and the monthly return autocorrelation of the real indices runs from 0.602 at Cleveland to 0.939 at Phoenix with a mean of 0.804. Every metro is more smoothed than the model of it, so the synthetic illiquid case of Section S29 was a conservative stand-in and not a flattering one. The second is coupling, which Section S14 found

orders the substitution rate at +0.92 across six equity targets. Here it spans 0.661 to 0.866 with a mean of 0.762, against 0.612 to 0.808 in Section S14. The two bands are of almost the same width, 0.205 against 0.196, which is the point: neither panel spans enough coupling to resolve a gradient, and the equity panel no longer spans more than this one does. That has one consequence in each direction and both were committed to before the rates existed. The level prediction is sharp and testable: every metro sits inside the equity coupling band or above it, none below its low end, with the panel's median in its upper half, so the mechanism predicts a high rate on all of them and no target on which substitution fails. The outcome is thirteen of fourteen on the variance coordinate, so the prediction is wrong once, at Las Vegas, and the subsection above records that the panel's coupling spread does not explain which metro it was wrong at. Thirteen of fourteen is the result, not fourteen of fourteen, and the difference is stated here because the prediction was committed to beforehand and a prediction that is only reported when it holds is not a prediction. The slope prediction is not testable here at all, and not because the equity panel spans more: it does not. Fourteen metros spanning 0.205 in coupling cannot resolve a gradient in a rate whose own interval is tens of points wide, and that is a band 1.04 times the width of the one that produced the equity ordering, which was itself read as a rank rather than a slope. Claiming the slope held on this panel would be the same error as claiming it failed.

Table S26. The panel, measured before any substitution rate was estimated. Persistence was computed for all twenty metropolitan indices and the persistence span quoted in the text is over all twenty; coupling was computed on the balanced panel of fourteen alone, so the coupling span is over the fourteen rows below, which are the metros with a full history from January 1987 and the metros the rates are estimated on. The level prediction is therefore committed to and tested on those fourteen, and nothing here says where the six later starters would have sat. The most persistent of the twenty is Phoenix at 0.939, which begins in 1989 and so is not a row here. Persistence is the lag-1 autocorrelation of monthly log returns over each metro's full history; Section S15 modelled a stale mark as partial adjustment at rho = 0.6, and every figure in that column exceeds it. Coupling is the correlation of the metro's own monthly return with the equal-weighted mean of its thirteen peers, the quantity Section S14 found orders the rate. The last two columns are the substitution rate at one quarterly appraisal cycle, reported here so the level prediction can be read across the panel rather than only in the aggregate.

metro	first mark	persistence	coupling	R(3), return	R(3), variance
Boston	1987-01	0.747	0.762	62.2%	47.9%
Charlotte	1987-01	0.713	0.699	67.4%	45.1%
Chicago	1987-01	0.761	0.748	98.0%	29.4%
Cleveland	1987-01	0.602	0.661	89.7%	9.7%
Denver	1987-01	0.789	0.704	36.5%	4.6%
Las Vegas	1987-01	0.859	0.710	28.2%	−11.4%
Los Angeles	1987-01	0.903	0.838	70.4%	32.0%
Miami	1987-01	0.872	0.769	76.6%	15.9%
New York	1987-01	0.843	0.735	75.5%	26.1%
Portland	1987-01	0.768	0.738	63.9%	30.6%
San Diego	1987-01	0.835	0.817	61.2%	9.6%
San Francisco	1987-01	0.834	0.821	63.1%	7.5%
Tampa	1987-01	0.834	0.807	91.4%	9.5%
Washington	1987-01	0.843	0.866	65.9%	6.3%

Table S27. Three peer blocks priced as a set, and the rule that picks one. $S_{cross}(0)$ is the cross arm's skill at zero delay, where the own mark is fresh and there is no substitution to detect, so it cannot favour a high rate; the bill is what the block costs before any substitution is available to pay for it. The selected row is the block with the highest $S_{cross}(0)$, differences below 0.01 treated as immaterial and ties broken towards the smaller block. On the variance target the widest block makes the rate negative at three of four delays and the compressed block makes it positive at all four.

peer block	cols	$S_{own}(0)$	$S_{cross}(0)$	bill	R(1)	R(2)	R(3)	R(6)
target: next month's return								
every peer separately	13	0.7242	0.7489	−0.0247	161.8%	91.3%	90.7%	88.3%
peer mean, one column ← **selected**	1	0.7242	0.7404	−0.0161	152.7%	66.2%	67.9%	69.1%
peer mean and its 3-month mean	2	0.7242	0.7458	−0.0216	138.3%	69.8%	72.2%	72.2%
target: next month's trailing twelve-month variance (backward-looking)								
every peer separately	13	0.9558	0.9162	+0.0396	−88.7%	−30.8%	−1.9%	26.1%
peer mean, one column ← **selected**	1	0.9558	0.9543	+0.0015	0.6%	10.2%	18.8%	34.8%
peer mean and its 3-month mean	2	0.9558	0.9556	+0.0002	20.6%	28.0%	30.5%	35.8%

Table S27 carries the most consequential finding in this section and it is methodological rather than substantive. On the variance target the widest block makes the substitution rate negative at one, two and three months, at −88.7%, −30.8% and −1.9%, and the compressed block makes it positive at every delay. The zero-delay column says why: thirteen peer columns cost 0.0396 of R^2 before any substitution is available to pay for them, and a rate computed against a cross arm already carrying a bill that size is measuring the bill. Compression leaves 0.0015 of it. Section S13's foreign-options comparison was confounded in the same way and corrected in the same way. The lesson to carry forward is that compression is part of the method and not a robustness check: at this sample size it is the difference between the central claim holding and appearing to fail.

The rule selected the one-column block on both targets, which was not built into it, and on the return target that choice cost real points. The widest block would have reported 90.7% at a quarterly staleness where the selected one reports 67.9%. What follows is the low end of the menu in Table S27 rather than a search over it.

The second objection is seasonality. These series are not seasonally adjusted and they share a large seasonal. Month-of-year explains 19.8% of monthly return variance on average and 41.7% at Cleveland, and the twelve-month profiles correlate +0.893 across metros on average. A target whose own mark is three months stale has lost its recent history, and a fresh peer mark tells it what month it is; if that were all the peers were doing, the rate would be measuring a calendar. Eleven month-of-year dummies enter both arms of every regression above, which closes the channel if the seasonal is a fixed additive monthly effect, and that is an assumption rather than a proof: the seasonal could drift, or be multiplicative, or the dummies could be estimated too noisily on 144 months to absorb it. Table S28 tests it with a placebo instead of another control, because a control can only fail to change the answer whereas a placebo can actively produce the wrong one and convict the design.

Table S28. The seasonal placebo. The real peer mean is replaced by a series carrying the shared seasonal and nothing else: the month-of-year mean of that peer mean, estimated on the training block alone and read off by calendar month, refreshed on the model's own twelve-month schedule. It has no dependence whatever on what any peer market did in the current month. The month-of-year dummies are removed from both arms in both columns, so the placebo is given every advantage the real series has.

delay	R, real peer mean	R, seasonal placebo
target: next month's return		
1 month	65.1%	−2.2%
2 months	68.9%	1.0%
3 months	72.9%	4.2%
6 months	70.6%	6.8%
target: next month's trailing twelve-month variance (backward-looking)		
1 month	−0.4%	−3.8%
2 months	9.3%	−0.8%
3 months	18.4%	−2.8%
6 months	34.5%	−4.0%

The placebo earns 4.2% where the real peer mean earns 72.9%, and on the variance target it earns −2.8% where the real mean earns 18.4%. So the rate is information and not a calendar effect, and that is established by a test that could have convicted the design and did not. Two smaller results sit alongside it. Removing the seasonal is the conservative choice on the return target rather than a favourable one: the untreated arm reports 72.9% against the published 67.9%, with causal deseasonalisation in between at 70.9%. And on the variance target the three treatments are indistinguishable at 18.8%, 15.9% and 18.4%, for a structural reason rather than an empirical one, since a twelve-month rolling variance necessarily spans every calendar month and a fixed month-of-year effect largely cancels inside the target before any model sees it.

Widening the cross-section from thirteen peers to nineteen, six of which begin between 1989 and 2000 and are therefore genuinely ragged, moves the rate from 67.9% to 67.5% on the return target and from 18.8% to 19.7% on the variance target. Nothing is lost, which is what compressing to a mean is for, and nothing is gained either, which is the more informative half. Six additional metros move the rate by less than a point, so thirteen peers already span whatever common factor the cross-section carries: that is the prediction of Section S27 and of the factor model in Section S26 holding on a second asset class, and it is of a piece with seasonal profiles that correlate +0.893 and couplings that sit in a band 0.2 wide. There is one national housing cycle in this panel, not nineteen independent signals.

S32. Robustness and scope: the remainder

Read as a clock, the trade has a price. A fifty-five-day-old domestic mark plus today's foreign closes forecasts as well as a plain domestic mark 4.6 days old [2.6, 8.2]. The same construction prices the other side of the trade: one day of staleness in the foreign feed costs 2.1 days [1.4, 3.2] of domestic freshness, and a feed one day behind turns the 51% recovered at five days into 29%, and the 36% at three days into 9%. Every rate in this paper assumes the cross-section arrives the same evening, and a cross-section older than the mark it replaces is worth less than nothing.

S33. Limitations: the four argued at length

The implied-volatility block is thinner than we would like. VIX and VDAX cover the sample; a spot VSTOXX index was obtained, tested and kept out of the headline block, because requiring it costs 73% of the sample, moving the window from 4,525 days beginning September 2008 to 1,240 beginning April 2020: the 2008 crisis is discarded outright and peak VIX falls from 82.7 over the full sample to 41.4 inside what remains. On the 1,240 days where all three exist, holding those days fixed so the third series is not confounded with the shorter and calmer window, it adds nothing at any delay and is mildly negative at all six: VSTOXX and VDAX are both European equity volatility indices and the extra column costs more to estimate than it returns. The conclusion a fuller block was meant to test is therefore unaffected. What remains open is the export rather than the conclusion, since it runs only from December 2012 to March 2025.

Two bounds on the tests themselves. The non-linearity result of Section S22 fits a random Fourier layer, which is a one-hidden-layer network whose first layer is drawn rather than learned; no network with a learned first layer was fitted, and the argument that one would not change the answer is an extension of the estimation-cost finding rather than a test of it. And Section S27's first principal component of seven equity indices is a proxy for global systemic risk, not a measurement of it, so the remainder it leaves may still carry global content this basis cannot see: the direction of that test is informative and its magnitude is not a bound. Section S28 names a limitation of a different kind, and it is the one a practitioner should read first. What breadth repairs is a stale forecaster's tendency to *over*-estimate variance, not their exposure to under-estimating it; the understating tail is unmoved at every delay we checked. Anyone reaching for the cross-section as protection against being caught short by a volatility spike is reaching for the wrong thing.

The largest of these limits is the estimator, and we can be specific about it rather than gesturing, because we fitted the alternative. Every model here is a ridge on a fixed feature set, chosen because it is transparent and because the object under study is the substitution rate rather than the best attainable forecast. The natural competitor is a one-factor state-space model with a Kalman filter that handles a missing observation by dropping that row from the measurement equation, which is what a delayed domestic mark *is* rather than something imposed on the data. Fitted two-step after Doz, Giannone and Reichlin (2011), with its parameters frozen at the training block, it beats this paper's specification at $\delta = 21$ and $\delta = 55$ and loses at $\delta = 0$, and the crossover is the interesting part: imposing a one-factor structure costs skill when the mark is fresh and pays when it is stale. Section S20 gives the intervals and the reconstruction diagnostics.

We report it rather than adopting it, and the reason is a scope judgement a reader is entitled to disagree with: $R(\delta)$ is a ratio in which the same domestic control appears three times, so a better estimator raises numerator and denominator together, and recomputing every conditional and every robustness check inside a filtered specification is a different paper. What matters for the headline is that the direction runs the opposite way to the usual limitation. Inside the filtered specification the rate at eleven weeks becomes 75.3% against the published 71.5%, a paired difference of +3.8% [+0.4%, +8.5%], and the mark reads 3.7 days old [2.0, 6.8] rather than 4.6. Those figures hold the filter's own parameters at values no forecaster had; re-estimated as the window grows the rate reads 72.6% and the advantage over the ridge no longer excludes zero, so what is established here is a direction and not a margin, and Section S20 gives both readings. A better estimator raises the numerator of $R(\delta)$ and has nothing to give its denominator, so this paper is not protecting a large

number with a transparent estimator; it is quoting a smaller one than the better estimator would give it. The filtered figure is the upper end of the 67.4% to 75.3% span the introduction reports across specifications, and the paper quotes the span rather than this figure alone.

S34. Further work: where the cross-asset test would break

The way it breaks is the more interesting outcome, and it is specific enough to look for. Commodity volatility has drivers that no equity cross-section can see: an inventory report, a harvest, a production decision by a cartel. If a commodity block adds where equity breadth does not, then "a global volatility factor seen early" is too coarse a description of what is happening, and the right cross-section depends on what the stale asset IS, and is not one panel for everyone. A farmland book and a private credit book would then need different peers, which is a materially different recommendation from the one Section S30 gives and would be worth the trouble of discovering.

Two warnings for whoever runs it. The models are the easy part; the data hygiene is not. Futures require contract rolls, so a continuous series carries a seam every quarter instead of the single seam per export audited in Section 4, and the audit, not the estimation, is where that work will actually go. And the admissibility table stops being a list of closing times: a bond future and its underlying cash market can trade on different calendars, so the rule that a peer counts only when its timestamp precedes the target's needs restating instrument by instrument instead of country by country.

S35. The admissibility rule on real session clocks

Section 3 states the rule the whole design rests on: a foreign close is admissible on day t only if its timestamp falls strictly before the target's close on that day. The rule is implemented from a table of fixed UTC closing times, and those are standard-time closes. Real exchanges keep local time, most of them observe daylight saving, and they do not switch on the same dates: the United States advances its clocks on the second Sunday in March and the European Union on the last Sunday, so for a fortnight each spring New York sits an hour closer to London than the table says. Until now the paper asserted that this never reorders a block. An assertion about a clock is a poor substitute for reading one, and this section reads them.

Every close in the panel is rebuilt in UTC from its exchange's own regular-session closing time in its own timezone, with the IANA database supplying every transition date, on all 6,991 dates in the sample. Two decisions are then compared, for every target, every peer and every date: the lag the fixed table assigns, and the lag the true closes imply. A disagreement in one direction is a *leak*, where the table calls a peer same-day admissible and the clock says it closed later or at the same moment, so the panel would carry a value the forecaster could not have seen. A disagreement in the other is *waste*, where the table sends a peer back a day that had in fact already closed.

There are **0** leaks. Across every target, every peer and every date, not one day on which the true session clocks would have reordered a block. The margin is not marginal either: the narrowest gap the rule anywhere relies on is **2.00 hours, between N225 and HSI**, so the fixed table is a simplification of the clock rather than an approximation of the rule. Daylight saving moves four of the eight closes by exactly one hour and moves none of them past a target.

The waste runs the other way and is worth stating because of its direction. On 3,336 of the 6,991 dates the table sends AXJO back a day for a N225 forecaster when the true clocks say it had already closed,

and 6,085 such days occur across the panel as a whole. The cause is that the table records the two markets at the same standard-time hour and the rule resolves equality against admissibility. So the implementation is *stricter* than the rule it implements: it discards information a real forecaster could lawfully have used. That direction is the one to prefer, and it also changes how the paper's numbers should be read. Every rate reported here is what a forecaster achieves with less than the admissible cross-section, which makes the measured substitution rate a lower bound on the rule's own terms rather than a figure flattered by a generous clock.

Holidays are the second assumption in the same family and they cannot leak, since a peer whose exchange was shut carries a close that is strictly older than the day it sits on. What they can do is weaken the instrument, so the frequency is reported rather than assumed: the peer carried forward most often is N225, on 5.8% of the target's days, and the mean age of a carried close is under three days. One parameter of that rule belongs with those figures, because a replicator needs it and the frequencies alone do not imply it: a peer is carried forward at most five trading days, after which its column is missing rather than stale, so no feature anywhere in this paper rests on a close older than a week. The foreign block is a block of recent closes, not a block of stale ones.

S36. The outage, split into information and estimation

Section 4.3 names what δ freezes and points here for the measurement. The training cut of equation (4) is forced by the vintage assumption: the label is the target's own decision variable, so a forecaster whose own mark is δ days old has not seen the outcomes that resolved since. The consequence, which an external audit put more sharply than an earlier objection had, is that δ does not isolate a stale input. It freezes the current domestic state, the labels available for fitting, the coefficient vintage, the ridge penalty the validation tail selects, and the quantity of training data, all at once. The design measures a domestic-data *outage*, and the question this section answers is how much of the damage belongs to each half.

The split needs an arm that no forecaster could run, and it is reported as such. Three arms share the feature construction, the estimator, the refit schedule and the test days, and differ only in where training stops.

Table S29. What δ costs, split. The feature-only arm holds the input δ days stale while cutting training at $t - h$, so its coefficients are as fresh as the estimator can make them. It is infeasible under this paper's own information set, since training to $t - h$ needs the stretch of the target's own series the delay withholds, and it exists to price one effect rather than to be reported as a forecast. Both other arms are the paper's.

δ	$S_{own}(0)$	feature only	outage	total damage	feature	vintage	vintage share
5	0.5188	0.3643	0.3637	0.1551	0.1545	0.0006	0.4%
13	0.5188	0.2343	0.2332	0.2856	0.2845	0.0011	0.4%
21	0.5188	0.1512	0.1478	0.3710	0.3676	0.0034	0.9%
55	0.5188	0.0209	0.0072	0.5116	0.4979	0.0137	2.7%

The answer is not close. Freezing the estimation accounts for **0.4%** of the delay's damage at one week and **2.7%** at eleven, and the two intermediate delays sit between them. What δ costs a forecaster is almost entirely the staleness of the input; the inability to re-estimate, re-tune and re-fit on more data is a rounding term against it. So the denominator $R(\delta)$ divides by is information, which is what the ratio was meant to be about.

The rate itself is recomputed inside the feature-only arm, with the cross-sectional model given the same cut so the comparison stays internal. The two rates differ by at most **2.8 points** across the grid, and the feature-only rate is the *higher* of the two at every delay, so the paper's design is also the conservative one: a forecaster who could keep re-fitting while their input went stale would find breadth repairing slightly more, not less.

What this settles is a naming question and not a result. The paper describes the experiment as an outage because that is what it is, and a reader who wants the share of stale-input damage that breadth repairs can take the feature-only column and will be within three points of the headline. What it does not settle is whether a real desk suffers outages or stale marks; that depends on the desk, and the two coincide for the case Section 1 opens with, where the mark and the data feed behind it stop together.

S37. The secondary grids, and what a reader may lean on · *Scope narrowed*

Section S6 prices the search behind the headline: nine specifications, a Reality Check and an SPA test, and the headline survives both. That covers the one comparison this paper was designed around and it covers nothing else. After the headline specification was fixed we went on to report several wide secondary grids, and they are enumerated here so a reader can add them up: eight markets by four delays on two blocks each for the compressed foreign block of Section S13, which is 64 cells; eight markets by four delays for the seven foreign closes given implied volatility, 32; six targets whose own index this paper does not hold, by four delays on two blocks each, 48; four nonlinear arms by six delays on two ridge grids, 48; and a third implied-volatility series at six delays, 6. The pool is 198, which is what the last row of the table counts. The enumeration used to sum to 102, because three of its five terms named a regressor count or an own-options count where the grid has a market count and two of them omitted the second block. Individual cells were marked significant inside them at the ordinary 5% level. None of those grids was pre-specified and no family-wise control was applied across them. A reader is entitled to know how many of those cells would look significant if every one were noise, and the answer is a number rather than a caveat.

Table S30. The post-selection audit. Each family is a grid reported after the headline specification was fixed. "Expected" is 0.05n, the count a grid of pure noise would show at the same threshold. The last two columns pool every secondary cell in the paper and apply a Bonferroni and a Benjamini-Hochberg correction across the pool; the headline comparison is excluded from the pool, because it carries its own correction in Section S6 and including it would let these grids dilute a test that was never part of them. Every count in this table is read from the labs' PRINTED tables, to two decimals, which is why its pooled count is 78 where the omnibus test below reads 77 from the stored series: three cells print a statistic equal to the critical value at that precision and cannot be classified from print. The two correction columns are computed from normal p-values, and the test below measures the null these statistics actually have, under which that reference distribution is wrong; they are reported as what the rule returns rather than as cells that have survived a correction.

family	cells	significant at 5%	expected by chance	survive BH	survive Bonferroni
compressed foreign block given implied volatility, eight markets	64	20	3.2	1	0
seven foreign closes given implied volatility, eight targets, two with their own options market	32	7	1.6	1	0
breadth given implied volatility, six targets whose own index is not held	48	18	2.4	3	2
nonlinearity given implied volatility, four arms	48	30	2.4	1	0
VSTOXX as a third implied-volatility series	6	3	0.3	0	0
pooled	**198**	**78**	**9.9**	**6**	**2**

Two readings are available and both are true. The first needs a null the column headed "expected" cannot supply: 0.05n is the right expectation and survives any dependence, but whether the observed count is surprising depends on how much that count VARIES, and these cells share evaluation days, training windows and nested delays. The omnibus test is run on the stored loss differentials rather than on the printed tables above, so its pool is the 198 recorded cells and its count is 77 of 198 cells significant under the table's own rule, the three print-boundary cells resolved by reading the series.

Resampling blocks of sixty days jointly within each of the twelve calendar groups, so that cross-cell dependence is carried into the null, the chance-only count has mean 18.3 and standard deviation 12.1 against a binomial 3.1, with a 95th percentile of 41. Seventy-seven is still well outside it: no replication of two thousand reached it, the largest being sixty-two, so the resampling p-value is at most 0.0005 – which is the smallest figure two thousand replications can return, a floor on the estimate rather than a ceiling on the truth, and zero exceedances in two thousand draws is consistent with a true p as large as 0.0015.

What the inflated null mean is, and what it is not. The mean of 18.3 is not cross-cell dependence carried into the null. Expectations are linear, so correlation between cells cannot move the mean of the count. It sits above 0.05n because the marginal rule is mis-sized on these series, rejecting 9.2% of the time rather than 5% under its own null, and the sum of the cells' individual rejection probabilities reproduces the null mean with no dependence at all. The worst cells are the random-Fourier-feature arms, whose loss differentials move in episodes on the scale of the resampling block; the rule's size there reaches a third. Dependence shows in the SPREAD, against independent cells at their own

sizes: 4.0 at the cells' own sizes against the observed 12.1, a factor of three, rather than the factor of four a binomial comparison suggests.

The rule that is correctly sized, and the margin it gives. Both rules above take a critical value from a reference distribution these statistics do not have. A third takes each cell's own 95% point of $|t|$ under the same resampling, calibrated on draws independent of those the count is judged against: its per-cell size is 5.2% and its null mean 10.3 against the 9.9 the table expects. Under it the observed count is 41 against a null mean of 10.3, with a 95th percentile of 25, and eleven of two thousand replications reached 41, for a resampling p-value of 0.0060. That is the margin this paper quotes. It is twelve times weaker than the bound the mis-sized rule returns, because a rule whose size is nearly double its nominal level inflates the observed count and the null mean together.

Two facts about the pool that a count does not convey. First, 32 of the 198 are a second copy of another cell: the seven-regressor arm is recorded once by the lab that runs it and once by the lab that compares it with a compressed one, with a byte-identical loss differential. On the 166 distinct cells the calibrated rule gives 35 observed against a null mean of 8.7 and a p-value of 0.0080, so the duplication is not what the result rests on. Second, the count is two-sided, and 50 of the 77 have a negative statistic: on these grids the added block more often significantly HURTS than helps. Under the calibrated rule it is 25 of 41. The test is direction-agnostic by design, since the question is whether the pool holds more structure than chance; what exceeds chance here is structure, not benefit.

The bandwidth the five labs pass is already scaled to each cell's delay; replacing the normal critical value with the fixed-b 95% point simulated at each cell's own length and bandwidth leaves seventy-one of the seventy-seven standing against a null 95th percentile of forty, under the same bound. That replacement changes little here because b is of the order of a hundredth in every long sample, where the fixed-b point is within its own simulation error of 1.96; it reaches 2.13 only in the six cells of the shortest series, and the correction matters exactly where the sample is short, which is the lesson of Section S12. The secondary grids collectively carry signal and are worth reporting, by a margin that is finite and, on the calibrated rule, modest. But only six of those seventy-eight survive a 5% false-discovery correction across the secondary set, and only two survive a family-wise one – both computed from normal p-values, which the measurement above shows to be the wrong reference distribution for these statistics, so neither survivor count is offered as a cell that has survived a correction. The seventy-two that do not are *exploratory*: real enough to tabulate, not firm enough to carry a claim on their own, and that is the word this paper uses for them from here on. Every table above whose cells fall in this audit is marked accordingly in its caption.

The instrument matters and it is chosen rather than defaulted to. Bonferroni controls the probability of any false rejection at all and is the right test if a single cell is to be leaned on; Benjamini-Hochberg controls the expected share of rejections that are false and is the right test for a grid read as a pattern, which is how these grids are actually read. We report both and lean on neither individually: the claim these grids support is the pattern, which survives, and not any one of its cells, which mostly does not.

Nothing here touches the headline, and the exclusion is deliberate rather than convenient. $\mathrm{R}(55)$ on the implementable target is a single pre-specified comparison, its multiplicity is priced in Section S6 against the nine specifications actually searched, and it survives Reality Check and SPA at five of six delays. Pooling it into a set of two hundred secondary cells would not be a stricter test of it; it would be a different and weaker one, in which a grid the headline was never part of decides the threshold the headline must clear.

S38. The volatility proxy, varied

Every number in this paper is computed on one measurement of the object being forecast: five-day Yang–Zhang on daily index open, high, low and close. The estimator is not varied anywhere else in the manuscript or in this appendix, and two things make that a sharper gap than the usual one-proxy objection.

The first is where the weight sits. At the five-day window used throughout, the weight k of equation (1) puts about an eighth of the estimator on the close term, a full unit on the overnight variance and the remainder on Rogers–Satchell – the constants are in Section 4.1 and are not restated here. The overnight term is built from the open and the previous close, and Rogers–Satchell from the open, the high, the low and the close, so the three fields a retail end-of-day export records least reliably, least synchronously and least well defined – the open, the high and the low – enter every term but the close-to-close one, which carries about an eighth of the estimator. Section S19 uses the high and the low entering Yang-Zhang and entering close-to-close variance nowhere as an audit instrument; it does not ask whether the proxy measures what it is meant to.

The second is mechanical, and it is the one that matters for this paper's reading. The overnight term is built from $\log(O_t/C_{t-1})$, and the interval between one New York close and the next New York open is exactly when Tokyo, Hong Kong, Mumbai, London and Frankfurt trade. The domestic feature therefore already contains, by construction, a reading of the global factor this paper identifies as the mechanism. That is not a leak – the clock audit of Section S35 is unaffected – but it means the published design cannot by itself separate *breadth substitutes for a stale mark* from *the overnight component of the target is a non-synchronous average of the same foreign sessions, and staleness destroys it*.

This section rebuilds the entire measurement – target, features, the 252-day median normalisation, the admissibility rule and the walk-forward – under four proxies rather than one, on the same test days, with the same control and the same inference. Two of the four contain no overnight term at all.

Table S31. The substitution rate at eleven weeks under four volatility proxies. CC uses the close alone and no open, high or low; RS and PK are intraday estimators and contain no overnight interval. Intervals are the fixed-b Fieller sets of Section 4.6 at the bandwidth and critical value stated there, and τ is the denominator's own t-statistic, the quantity Section 4.6 shows must exceed 2.11 for the set to be bounded. $S_{\text{own}}(0)$ is the domestic model's skill with nothing withheld, reported because it is the anchor the rate divides against.

proxy	built from	$S_{\text{own}}(0)$	R(55)	fixed-b 95% set	τ
YZ	the published estimator	0.5188	71.5%	[54.3%, 83.4%]	5.40
CC	close-to-close; no open, high or low	0.3037	73.4%	[47.5%, 89.4%]	4.32
RS	Rogers–Satchell; no overnight term	0.4684	69.4%	[50.5%, 82.9%]	5.33
PK	Parkinson; no overnight term	0.5455	70.1%	[51.5%, 84.8%]	5.56

The rate at eleven weeks reads 71.5% on the published estimator, 69.4% and 70.1% on the two that remove the overnight term entirely, and 73.4% on close-to-close, which uses no open, high or low at all. Every set excludes zero and every denominator is identified. So the explanation this section was written to test – that the cross-section is predicting a component of the target which is itself a measurement of those same foreign sessions – does not survive: removing that component moves the rate by about two points, and the arm built from the one field a retail export records reliably returns the highest rate of the four.

The result is stronger than a stability claim, because the denominator's anchor is not stable. $S_{\text{own}}(0)$ runs from 0.3037 on close-to-close to 0.5455 on Parkinson – the domestic model forecasts some of these targets far better than others – and the ratio is nevertheless within a few points of itself throughout. A quantity that held only because its parts held would be a weaker finding than one that holds while its parts move.

Two qualifications belong with it. The arms disagree at short delays, and widely: at five days the rate reads 51.4% on Yang–Zhang against 92.2% on close-to-close. That is not a contradiction of the above but an instance of it, because five days is where Section 4.6 says the denominator is small and the ratio weakly identified, and a quantity the data does not pin down is free to move with the proxy. The agreement at eleven weeks and the disagreement at five are the same fact about identification, seen twice. And this is one measurement varied, not a demonstration that the target is well measured: no intraday data is used anywhere in this replication, and a realised-variance target built from five-minute returns would be a better object than any of the four tested here. That is the first thing we would add with access to intraday data, and Section S34 records it as such.

S39. What a conditional set covers, at its own length

Section 4.6 prints τ beside every interval in this appendix so a reader can see which cells are weakly identified. That is necessary and it is not sufficient: τ says whether the set is bounded, and says nothing about what a bounded set covers. This section supplies the missing column, because two coverage studies already here answer neighbouring questions and neither answers this one.

The coverage study in Section 4.6 of the paper sweeps the identification strength at the full evaluation length and reports both critical values; its normal-critical-value arm is the procedure this paper replaced. Section S10's ladder sweeps the length at one strength with the fixed-b value, and is what established that a 367-day slice covers 65% where the full window covers 95%. The cells of Table S4 are neither: fixed-b sets, like S10's, but computed on about a third of the sample at strengths between 2.9 and 5.8. Reading S10's ladder across to them would put their coverage near the 85% it records at 866 days. That inference is available to a reader and it is wrong, which is the reason to measure rather than interpolate.

Table S32. Simulated coverage of the nominal 95% fixed-*b* Fieller set at each Table S4 cell's own length and own identification strength, against the generator of Section 4.6 matched to the measured loss-differential autocorrelation. 1,500 replications per row; the Monte Carlo error on a coverage figure near 90% is about 0.8 points. The last column is the share of replications in which the set is not an ordinary interval, which is the pathology Section 4.6 describes and the calm cells display.

state	δ	*n*	τ	covers	unbounded
calm	5	1,620	3.4	93.5%	6.5%
calm	13	1,620	2.9	92.8%	14.8%
calm	21	1,620	3.5	92.4%	5.9%
calm	55	1,620	3.0	93.7%	12.1%
middle	5	1,624	5.6	93.3%	0.0%
middle	13	1,624	5.8	94.1%	0.1%
middle	21	1,624	5.5	92.7%	0.0%
middle	55	1,624	4.4	94.3%	0.7%
stressed	5	1,618	3.5	93.5%	5.2%
stressed	13	1,618	3.7	92.8%	3.6%
stressed	21	1,618	3.6	93.0%	4.7%
stressed	55	1,618	4.2	93.1%	1.1%
full window	*55*	*4,862*	*5.4*	*94.3%*	*0.1%*

Across the twelve conditional cells the nominal 95% set covers between 92.4% and 94.3%. The cell Section 3 tells a practitioner to carry, stressed at eleven weeks, covers 93.1%, against 94.3% for the same procedure on the whole sample. The conditional sets are therefore worth about a point less than the unconditional one, not the seven or eight points S10's length ladder would suggest, and the reason is that the ladder's short rungs are weakly identified as well as short while these cells are not: none of them is below $\tau = 2.9$.

Two things in the table are worth more than the headline. The calm cells return a set that is not an ordinary interval in 14.8% and 12.1% of replications at thirteen and fifty-five days, against 0.7% in the middle tercile and 1.1% under stress, so the calm cell's refusal to produce a usable number is reproduced by the simulation rather than merely asserted from the data. And the shortfall from the nominal level is a property of length rather than of identification: τ in the middle tercile is higher than in the full window at three of its four delays, and all three of those still cover less.

These are coverage figures for this procedure against one model of the dependence, matched to one measured autocorrelation at one delay. They are not confidence statements about the cells themselves, and a reader should treat them as the instrument's calibration rather than as a second interval. They are reported for the same reason τ is: someone deciding how hard to lean on a conditional number is entitled to know what the instrument does, and the honest answer is not the nominal level.

S40. The scope condition, tested within target

Section 7 concludes that where a liquid options market exists on the asset itself, breadth is redundant. Until now that rested on two assets, the S&P and the DAX, because they were the only targets for

which this paper held an own-index volatility index. Section S13 could do no better than set them against six targets holding somebody else's, and its own Part D concedes the difficulty: across those eight, option-market status and coupling to the cross-section are perfectly separated, every optioned target being more strongly coupled than every unoptioned one. With eight points the two cannot be told apart, so the gradient reported there is equally consistent with *options make breadth redundant* and with *well-coupled markets are the ones that happened to get options*.

This section breaks that confound by never comparing two targets. For each of four markets it runs the same target, over the same window, against the same cross-section, and changes one thing: whether the implied-volatility block is written on the asset being forecast or on somebody else's. Write A(T) for what the foreign cross-section adds to a forecaster holding foreign implied volatility and B(T) for what it adds to one holding their own. Coupling, clock, sample, estimator and peer block are identical between the two arms because it is one target; only the options feed moves.

The four series are the Nikkei Stock Average Volatility Index for N225, India VIX for NSEI, the HSI Volatility Index for HSI and the S&P/ASX 200 VIX for AXJO. A published history can begin *earlier than the index existed*. Nikkei VI began real-time calculation on 19 November 2010 and is back-calculated to 1989, and VHSI launched on 21 February 2011. A back-calculated value was never on anyone's screen, so feeding one to a forecaster is the same error as a stale mark running backwards. The exports used here begin at each index's live date and carry no back-cast rows, which `lab73` checks rather than assumes. Each target's two arms are estimated on that target's own-IV window and are therefore length-matched to each other, at 2,052, 2,746, 2,014 and 2,915 test days; they are shorter than the headline test and are not comparable to it.

An own-index volatility index prints when its own cash market closes, which is the VIX-against-S&P problem of Section S28 and is handled the same way: both arms are reported, one with the own index entering at t and one at $t-1$, the second conceding the whole session as a conservative bound. The verdict is the same at both ends of that interval.

Table S33. What the foreign cross-section still adds at eleven weeks, given foreign implied volatility (A) and given the target's own (B). Same target, same rows, same peers, same estimator; only the options feed differs. Giacomini–White, since the model without the peer block is nested in the model with it.

target	own index	test days	A gain	GW z	B gain	GW z
N225	NKVI	2,052	−0.0126	−0.69	−0.0427	−2.86
NSEI	INDIAVIX	2,746	+0.0067	+0.31	+0.0033	+0.14
HSI	VHSI	2,014	−0.0143	−0.59	−0.0173	−1.58
AXJO	AVIX	2,915	+0.0108	+0.54	+0.0068	+0.41

Breadth is worth less once the options are the asset's own on four of four targets, and on four of four again in the strict arm, where the same differences read −0.0279, −0.0005, −0.0046 and −0.0014. The one cell that clears a statistic of two is N225 given its own options, where breadth does not merely fail to help but costs 0.0427. So the scope condition of Section 7 now holds on six markets rather than two.

Two things this does not establish, and the second matters more than the first. Four targets agreeing on a sign is worth reporting and is not worth an interval, and none is quoted: three of the four differences are between −0.003 and −0.004, which is noise. And the paired design does **not** separate options from coupling, because breadth adds nothing to these four *either way*: A is significant in zero of four as

well. There is no collapse to observe when the starting value was already zero. Separating the two would need a target on which breadth genuinely helps given foreign options, and in this panel that means the frontier markets of Section S14, which have no own volatility index to hold. That limitation is structural and more data of this kind does not fix it.

The result worth carrying out of this section is not the difference but the level. At eleven weeks the own-index block scores 0.3191 against 0.0749 for the foreign block on N225, 0.4748 against 0.1766 on HSI, 0.4576 against 0.2583 on NSEI and 0.4220 against 0.3199 on AXJO: 1.3 to 4.3 times as high, on every target, at every delay. Where an options market exists on the thing being forecast, nothing else measured in this paper comes close to it, and that is a sharper statement of Section 7's practical advice than Section 7 makes.

One provenance note belongs here rather than only in the data README. A-VIX is taken from Yahoo Finance and not from the vendor behind the other three, because that vendor's export carries 75 rows dated on a Saturday or a Sunday in blocks running from early October to late March (the Australian daylight-saving window, which is a Sydney timestamp converted without adjusting the date), together with a print of 101.00 on Sunday 23 May 2010 between a Friday close of 33.22 and a Monday close of 30.88. The two vendors were compared rather than one preferred on faith: on the 4,743 weekday dates both cover they agree to within 0.5% on 99.92%, at a median absolute difference of 0.0020. This is one series cross-validated across two independent vendors, with one vendor's copy rejected for misdating and a bad print, not a change of series.

S41. Whether any input is dated on a day its market was shut

This paper is careful about time. Every series carries a close in UTC, a peer closing at or after the target is shifted back a day, implied volatility enters at $t-1$ where its print is stamped after the target's, `CONVENTIONS.tsv` records the rule for each series and Section S35 audits the session timestamps. All of that machinery compares one timestamp with another. None of it ever compared a timestamp with a *calendar*.

So a row dated on a Sunday passed every check in the project. It has a date, the date is in range, it sorts correctly, it precedes the target's close, and no digest notices, because a digest is over the bytes the vendor shipped and the vendor shipped exactly those bytes. The row is simply not a cash print of the index whose column it sits in, and nothing here could see that.

Three index series carried such rows, and two of the three were wrong. They are told apart by comparing each series with *itself* on two properties a cash close and a futures print do not share: how often the close is a whole number, and the mean intraday range. A cash index is a weighted sum of constituent prices and lands on a round figure by accident; a futures contract trades in ticks and prints round numbers often. N225's thirty weekend rows, all between 25 January and 6 September 2026, are whole numbers 66.7% of the time against 1.0% of its weekday rows, with a mean (high − low)/close of 0.239% against 1.099%. They are Nikkei *futures*, which trade through the weekend, pasted into a cash export. HSI's single row, 30 November 2008, prints 17,572.00 with the cash index at 13,888.24 on the Friday and 14,108.84 on the Monday, a 26% error. NSEI's twenty-nine are 3.4% whole numbers against 5.4% on weekdays with a 1.022% range against 1.596%, which is an ordinary cash session. They are genuine: the National Stock Exchange runs Saturday budget-day, Muhurat and disaster-recovery sessions, and India VIX, computed from Nifty *options* rather than from the cash market, independently prints on exactly the same Saturdays.

A single wrong print is not a single wrong number. Yang–Zhang's overnight term is $\log(O_t \,/\, C_{t-1})$, so a spurious row corrupts its own proxy value and the following day's, and the decision variable then divides by a trailing 252-day median which carries the damage forward for a year of trading days. HSI's one bad row corrupted 144 rows of that target's decision variable, 2.18% of its panel; N225's thirty corrupted 162, or 2.48%.

The rule is now part of the design: a row is admissible only on a day its own exchange trades, with exchanges that hold weekend sessions named explicitly, and it is applied before the volatility proxy is computed so the overnight chain reconnects across the gap rather than inheriting it. The vendor files are untouched, because a manifest digest is a statement about the bytes the vendor shipped and editing one to repair a vendor error would break that chain and bury the defect; the rule lives in the code that interprets them and is written into `CONVENTIONS.tsv` from the gate itself.

What it changed. Nothing this paper claims, and the last decimal of most of what it reports. The headline rate moves from 71.6% to 71.5% and its interval from [54.4, 83.5] to [54.3, 83.4], with the identification statistic unmoved at 5.40. The multi-target tables of Sections S13 and S14 move by up to 1.3 points on Hong Kong, whose own series carried the bad print, and the counts those sections turn on (nought of eight, one of twenty-four, three of sixteen, eight of sixteen) are all unchanged. The housing study of Section S31 is byte-identical, since it shares no series with the equity panel. The honest summary is that this defect cost the paper no conclusion and should still never have survived two years of checking, and that it did so because every check here compared a timestamp with another timestamp, and a calendar is a different kind of fact.